

Chabauty Techniques in Number Theory



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A dissertation submitted for Master's of Mathematics

University of Oxford

Trinity Term, 2022

There is one comforting conclusion which is easy for a real mathematician. Real mathematics has no effects on war. No one has yet discovered any warlike purpose to be served by the theory of numbers or relativity, and it seems very unlikely that anyone will do so for many years.

— G. H. Hardy

A Mathematicians Apology (1940)

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Notation

1. Δ_F denotes the discriminant of a polynomial F .
2. $R[[X]]$ denotes the ring of power series with coefficients from a commutative ring R .
3. C_p denotes the cyclic group of order p .
4. $G[2]$ denotes the 2-torsion subgroup of a group G .
5. $[G]^{\times n}$ denotes the direct product of n copies of a group G .

6. $[2]_G$ denotes the multiplication by 2 map on a group G .
7. $\mathcal{C}(\mathbb{Q})$ denotes the set of $\mathbb{Q}$ -rational points on a curve $\mathcal{C}$.

0 Introduction

Throughout this paper, K will be a field, usually $\mathbb{Q}$ or a number field. Occasionally, software packages are needed to carry out computations, the systems used are Magma (<http://magma.maths.usyd.edu.au/calc/>) and SageMath (<https://sagemath.org/>).

From the 3rd century, Problem VI.17 of the *Arithmetica*, to a more recent development on Fermat equations, the theory of Chabauty techniques always, strikingly, played a vital role in making a progress.

Theorem 0.1 (Chabauty’s Theorem). Let C be a curve over a number field K with genus g . Suppose its Jacobian has rank $\leq g-1$, then C has finitely many K -rational points.

Despite this being weaker than Falting’s Theorem, an advantage of Chabauty’s Theorem is that it gives a bound on the number of K -rational points when applicable. We will therefore explore Chabauty techniques in this paper and apply it in different situations.

This paper will be divided into three sections. The first will be devoted to curves of genus 1, by which we will focus on elliptic curves. We then will investigate Chabauty techniques on elliptic curves over a number field. The second section will focus on curves of genus 2, with the treatment being similar to the first section; we will first construct an Abelian group associated to it and investigate its structure, followed by finding $\mathbb{Q}$ -rational points using classical Chabauty. Finally, we will conclude this paper by considering problems where the aforementioned techniques could not be directly used but will nonetheless provide an excellent tool for solving the problems.

It is intended that the readers are familiar with B3.1 Galois Theory, B3.4 Algebraic Number Theory and C3.7 Elliptic Curves.

1 Group Structure on Elliptic Curves

To start off, we will give a brief overview of C3.7 Elliptic Curves. This will be followed by considering the rank of an elliptic curve over a number field, using a non-homogeneous approach.

Definition 1.1. An **elliptic curve** over K is, up to birational equivalence, a non-singular projective cubic curve with a K -rational point.

Theorem 1.2. Let $\mathcal{E}$ be an elliptic curve over K . The points on $\mathcal{E}$ form an Abelian group. Moreover, the K -rational points on $\mathcal{E}$ form a subgroup, the **Mordell-Weil**

group, denoted by $\mathcal{E}(K)$.

Comment 1.3. In practice, the form of an elliptic curve will be $Y^2 = F(X) = X^3 + AX^2 + BX + C$, where $A, B, C \in K$. We choose the point at infinity, denoted by ∞ , to be the identity of the group. We further denote $\mathcal{E}(K)_{\text{tors}}$ to be the torsion subgroup of $\mathcal{E}(K)$.

Comment 1.4. An elliptic curve over $\mathbb{Q}$ is birationally equivalent to $Y^2 = X^3 + aX^2 + bX + c$ where $a, b, c \in \mathbb{Z}$.

1.1 Elliptic Curves over $\mathbb{Q}$

1.1.1 Mordell-Weil group

Theorem 1.5. Suppose $\mathcal{E} : Y^2 = F(X) = X^3 + aX^2 + bX + c$ where $a, b, c \in \mathbb{Z}$. Let p be a rational prime such that $p \nmid 2\Delta_F$, then $\mathcal{E}(\mathbb{Q})_{\text{tors}} \hookrightarrow \tilde{\mathcal{E}}(\mathbb{F}_p)$, where we denote $\tilde{\mathcal{E}}$ to be the reduction of the elliptic curve $\mathcal{E}$ modulo p .

Theorem 1.6 (Mordell-Weil Theorem). Let $\mathcal{E}$ be an elliptic curve over $\mathbb{Q}$, then $\mathcal{E}(\mathbb{Q})$ is finitely generated.

Corollary 1.7. $\mathcal{E}(\mathbb{Q}) \cong \mathcal{E}(\mathbb{Q})_{\text{tors}} \times \mathbb{Z}^r; r \in \mathbb{N}_0$. r is the **rank** of the elliptic curve.

1.1.2 Rank computation via 2 isogeny

Let $\mathcal{C}$ be an elliptic curve over $\mathbb{Q}$ with a $\mathbb{Q}$ -rational order 2 point; using a birational transformation if necessary, we may assume this point is $(0, 0)$. Thus it suffices to consider the elliptic curve of the form $\mathcal{C} : Y^2 = X(X^2 + aX + b)$.

Lemma 1.8. Consider the elliptic curve $\mathcal{D} : V^2 = U(U^2 + \hat{a}U + \hat{b})$ with $\hat{a} = -2a$, $\hat{b} = a^2 - 4b$. There exists 2-isogenies $\phi : \mathcal{C} \rightarrow \mathcal{D}, \hat{\phi} : \mathcal{D} \rightarrow \mathcal{C}$ which induces group homomorphisms between the relevant Mordell-Weil groups. Moreover, $\phi \circ \hat{\phi} = [2]_{\mathcal{D}(\mathbb{Q})}$ $\hat{\phi} \circ \phi = [2]_{\mathcal{C}(\mathbb{Q})}$.

Furthermore, with some abuse of notation, define $q : \mathcal{D}(\mathbb{Q})/\phi(\mathcal{C}(\mathbb{Q})) \rightarrow \mathbb{Q}^*/(\mathbb{Q}^*)^2$ by

$$q((u, v) + \phi(\mathcal{C}(\mathbb{Q}))) = \begin{cases} u & u \neq 0 \\ \hat{b} & u = 0 \\ 1 & (u, v) = \infty. \end{cases}$$

Analogously, define $\hat{q} : \mathcal{C}(\mathbb{Q})/\hat{\phi}(\mathcal{D}(\mathbb{Q})) \rightarrow \mathbb{Q}^*/(\mathbb{Q}^*)^2$ by

$$\hat{q}((x, y) + \hat{\phi}(\mathcal{D}(\mathbb{Q}))) = \begin{cases} x & x \neq 0 \\ b & x = 0 \\ 1 & (x, y) = \infty. \end{cases}$$

These maps are injective group homomorphisms and the images are finite.

Comment 1.9. In practice, by using homogeneous spaces¹, we could find the relevant images and therefore construct $\mathcal{C}(\mathbb{Q})/2\mathcal{C}(\mathbb{Q})$. Notice that $\mathcal{C}(\mathbb{Q})/2\mathcal{C}(\mathbb{Q}) \cong \mathcal{C}(\mathbb{Q})[2] \times [C_2]^{\times r}$, so the rank could be calculated once $\mathcal{C}(\mathbb{Q})[2]$ is found. This, together with Theorem 1.5, give an algorithmic method of determining the Mordell-Weil group for an elliptic curve over $\mathbb{Q}$, at least in practice.

1.2 Elliptic Curve over a Number Field

For this section, let $K : \mathbb{Q}$ be a number field and $\mathcal{E}$ be an elliptic curve over K . As usual, $\mathcal{O}_K$ denotes the ring of integers of K , the subscript is often dropped if the context is clear. We also denote $K_{\mathfrak{p}}$ to be completion of K at $\mathfrak{p}$ (see Appendix 6.1 for more detail), where $\mathfrak{p}$ is a prime ideal of $\mathcal{O}$.

1.2.1 Mordell-Weil group

Theorem 1.10. [Bru99, p. 14] Suppose $\mathcal{E} : Y^2 = X^3 + AX^2 + BX + C$ is an elliptic curve over K . Suppose further that $\mathfrak{p}$ is a good reduction² and $\nu_{\mathfrak{p}}(\text{char}(\mathfrak{p})) < \text{char}(\mathfrak{p}) - 1$, then³

$$\mathcal{E}(K)_{\text{tors}} \hookrightarrow \tilde{\mathcal{E}}(\mathcal{O}/\mathfrak{p}).$$

Theorem 1.11 (Mordell-Weil Theorem). Let $\mathcal{E}$ be an elliptic curve over K , then $\mathcal{E}(K)$ is finitely generated. Thus $\mathcal{E}(K) \cong \mathcal{E}(K)_{\text{tors}} \times \mathbb{Z}^r; r \in \mathbb{N}_0$. r is the **rank** of $\mathcal{E}$.

Proof. See [Sil09, p. 239], Theorem 6.7. □

Again, $\mathcal{E}(K)/2\mathcal{E}(K) \cong \mathcal{E}(K)[2] \times [C_2]^{\times r}$. This isomorphism will aid the rank computation.

1.2.2 Rank computation via complete 2 descent

The theory of 2-isogeny could still be applied to find the rank of $\mathcal{E}$; Section 4.3 of [Bru99] describes a detailed approach, which generalises the ideas in 1.1.2. We will, however, describe the method of complete 2 descent.

Suppose our elliptic curve $\mathcal{E}$ over K is of the form:

$$\mathcal{E} : Y^2 = F(X) = (X - x_1)(X - x_2)(X - x_3).$$

¹See C3.7 Elliptic Curves Lecture Notes. Available at https://courses.maths.ox.ac.uk/pluginfile.php/22127/mod_resource/content/1/EC_notes_HT2022.pdf.

²See [Bru99] Section 2.3 for a notion of a *good reduction*. Usually $\mathfrak{p}$ is a good reduction if it does not divide the discriminant in $\mathcal{O}_K$.

³See Appendix 6.1 for the definition of $\nu_{\mathfrak{p}}$.

Lemma 1.12. Consider the following map: $\mu: \mathcal{E}(K)/2\mathcal{E}(K) \rightarrow [K^*/(K^*)^2]^{\times 2}$

$$P \mapsto \begin{cases} [1, 1] & P = \infty \\ [(x_1 - x_2)(x_1 - x_3), x_1 - x_2] & P = (x_1, 0) \\ [(x_2 - x_1), (x_2 - x_1)(x_2 - x_3)] & P = (x_2, 0) \\ [x - x_1, x - x_2] & P = (x, y), P \neq (x_1, 0), P \neq (x_2, 0). \end{cases}$$

μ defines an injective group homomorphism and the image of μ is finite.

Proof. See Theorem 3 of [Lan] for a proof via Galois cohomology. $\square$

Comment 1.13. As stated in Theorem 3 of [Lan], Lemma 1.12 holds for any field of characteristic of 0, not just a number field.

To determine $\text{Im}\mu$, we consider [Figure 1], where $i_{\mathfrak{p}}$ and $j_{\mathfrak{p}}$ are the canonical inclusions and $\mu_{\mathfrak{p}}$ is the natural extension of μ . As stated in Comment 1.13, $\mu_{\mathfrak{p}}$ is still an injective group homomorphism. Let M be a finite group containing $\text{Im}\mu$. For example, let S be the set of places of bad reductions union 2 and let $\mathcal{S}$ be the set of primes lying above the elements of S ; then M can be taken to be the set consisting the elements of $[d_1, d_2] \in [K^*/(K^*)^2]^{\times 2}$ where $K(\sqrt{d_1}) : K$ and $K(\sqrt{d_2}) : K$ are unramified at primes not in $\mathcal{S}$. Let $M_{\mathfrak{p}}$ be the $\mathfrak{p}$ -adic equivalent of M . [FR03]

$$\begin{array}{ccc} \mathcal{E}(K)/2\mathcal{E}(K) & \xrightarrow{\mu} & M \\ i_{\mathfrak{p}} \downarrow & & \downarrow j_{\mathfrak{p}} \\ \mathcal{E}(K_{\mathfrak{p}})/2\mathcal{E}(K_{\mathfrak{p}}) & \xrightarrow{\mu_{\mathfrak{p}}} & M_{\mathfrak{p}} \end{array}$$

Figure 1: Complete 2 descent - elliptic curves

Comment 1.14. Suppose $\mathcal{S} = \{p_1, \dots, p_m\}$ and let $K(\mathcal{S}) = \{up_1^{r_1} \dots p_m^{r_m} : u \in \mathcal{O}^{\times}, r_i = 0, 1\}$. $K(\mathcal{S})$ may be perceived to be a subgroup of $K^*/(K^*)^2$. If $\mathcal{O}^{\times}$ is finite then one can take $M = [K(\mathcal{S})]^{\times 2}$.

Lemma 1.15. The above is a commutative diagram and so $\text{Im}\mu \subseteq j_{\mathfrak{p}}^{-1}(\text{Im}(\mu_{\mathfrak{p}}))$.

Proof. Clear. $\square$

Comment 1.16. Using Lemma 1.15, we have that $\text{Im}\mu \subseteq \bigcap_{\mathfrak{p}} j_{\mathfrak{p}}^{-1}(\text{Im}(\mu_{\mathfrak{p}}))$ and so this provides an upper bound on $\text{Im}\mu$. $\bigcap_{\mathfrak{p}} j_{\mathfrak{p}}^{-1}(\text{Im}(\mu_{\mathfrak{p}}))$ is known as the 2-*Selmer group*. Whilst intersecting all possible $\mathfrak{p}$ seems cumbersome, Proposition 4 of [Lan] shows that it is sufficient to intersect primes in $\mathcal{S}$ and primes at infinity of K .

Suppose H is a known subgroup of $\text{Im}\mu$ and is suspected to be $\text{Im}\mu$. One may hope $H = \bigcap_{\mathfrak{p}} j_{\mathfrak{p}}^{-1}(\text{Im}(\mu_{\mathfrak{p}}))$ in order to confirm that $H = \text{Im}\mu$. In practice, one usually takes H to be the subgroup generated by the known images of μ .

Since [Figure 1] concerns about $\mathcal{E}(K_{\mathfrak{p}})/2\mathcal{E}(K_{\mathfrak{p}})$, the following lemma helps to determine if a set of elements could represent $\mathcal{E}(K_{\mathfrak{p}})/2\mathcal{E}(K_{\mathfrak{p}})$.

Lemma 1.17. Let K be a number field and $\mathcal{E}$ be an elliptic curve over K . Let $\mathfrak{p}$ be a prime of K , then

$$\#\mathcal{E}(K_{\mathfrak{p}})/2\mathcal{E}(K_{\mathfrak{p}}) = \#\mathcal{E}(K_{\mathfrak{p}})[2] \times \#\mathfrak{p}/2\mathfrak{p}.$$

Proof. See [CF96, p. 72] for a proof. $\square$

The theory described above, with a bit of luck, is enough to determine the Mordell-Weil group. The following original example on rank computation is heavily inspired by Lemma 5 of [FR03].

Example 1.18. Let $K = \mathbb{Q}(i)$ and $\mathcal{E}_1 : Y^2 = F(X) = (X - 2)(X + 2)(X + 2i)$ over K . $\mathcal{E}_1$ has rank 1.

Proof. Trivially, $\mathcal{E}_1(K)[2] \cong [C_2]^{\times 2}$ since $F(X)$ fully factorises. In fact, Magma shows that $\mathcal{E}_1(K)_{\text{tors}} \cong \mathbb{Z}/2 \times \mathbb{Z}/2$. Moreover, $(0, 2i - 2) \in \mathcal{E}(K)$ is torsion free since the torsion subgroup is known. Simple calculations show that the primes of bad reduction coincide with primes lying above 2 since the discriminant is 1024. By Comment 1.14 we take $M = \langle [i, 1], [1 + i, 1], [1, i], [1, 1 + i] \rangle$.

Let $A = (2, 0), B = (-2, 0), C = (-2i, 0)$ and $D = (0, 2i - 2)$. Under the map μ described in Lemma 1.12, we see that:

$$\begin{aligned} A &\mapsto [(1 + i)i, 1] \\ B &\mapsto [1, 1 + i] \\ D &\mapsto [i, i]. \end{aligned}$$

Notice that these elements are independent in $K^*/(K^*)^2$. Let $H = \langle [(1 + i)i, 1], [1, 1 + i], [i, i] \rangle$, we wish to show that $H = \bigcap_{\mathfrak{p}} j_{\mathfrak{p}}^{-1}(\text{Im}(\mu_{\mathfrak{p}}))$.

$\mathfrak{p} = \infty$ gives no information because $\mathbb{Q}(i)_{\infty} = \mathbb{C}$ and $\mathbb{C}^*/(\mathbb{C}^*)^2$ is the trivial group. Thus it remains to just check $\mathfrak{p} = 1 + i$, by Comment 1.16.

Firstly, it is clear that $\ker(j_{1+i})$ is the trivial subgroup of M and that

$$\begin{aligned} A &\mapsto [(1 + i)i, 1] \\ B &\mapsto [1, 1 + i] \\ D &\mapsto [i, i] \end{aligned}$$

under μ_{1+i} .

These elements remain to be independent in $\mathbb{Q}(i)_{1+i}^*/(\mathbb{Q}(i)_{1+i}^*)^2$. In the light of Lemma 1.17, there is exactly one generator missing for $\mathcal{E}_1(\mathbb{Q}(i)_{1+i})/2\mathcal{E}_1(\mathbb{Q}(i)_{1+i})$. After a short search, one finds that $(1 + 2i, \gamma) \in \mathcal{E}_1(\mathbb{Q}(i)_{1+i})$ and is mapped to

$[-1+2i, 3+2i]$. Such γ exists because $(1+2i-2)(1+2i+2)(1+2i+2i) = -23-24i$ is a square in $\mathbb{Q}(i)_{1+i}^*$, by Hensel's Lemma if necessary. Since, $\{[-1+2i, 3+2i], [(1+i)i, 1], [1, 1+i], [i, i]\}$ forms a set of independent elements, $\mathcal{E}_1(\mathbb{Q}(i)_{1+i})/2\mathcal{E}_1(\mathbb{Q}(i)_{1+i}) = \langle A, B, D, (1+2i, \gamma) \rangle$. However, there are no elements in M that can be mapped to $\langle [-1+2i, 3+2i] \rangle$ except the identity element. Thus, $j_{1+i}^{-1}(\text{Im}(\mu_{1+i})) = H$ and so $H = \bigcap_{\mathfrak{p}} j_{\mathfrak{p}}^{-1}(\text{Im}(\mu_{\mathfrak{p}}))$ is indeed true.

Since $H = \text{Im}\mu \cong [C_2]^{\times 3} \cong \mathcal{E}_1(K)[2] \times [C_2]^{\times r}$ and $\mathcal{E}_1(K)[2] \cong [C_2]^{\times 2}$, one deduces the rank is indeed 1. $\square$

2 Chabauty Techniques on Elliptic Curves

Let K be a number field. Previously, we have seen that $\mathcal{E}(K)$ could be calculated. This chapter will be dedicated to bounding the number of K -rational points on $\mathcal{E}$ with x or y -coordinate being $\mathbb{Q}$ -rational using $\mathcal{E}(K)$. We shall first recall a few elementary results.

Assume $\mathcal{E}$, an elliptic curve over K , has the following form

$$\mathcal{E} : Y^2 = g_3X^3 + g_2X^2 + g_1X + g_0. \quad (2.1)$$

2.1 Preliminary Results

Suppose $(x, y) \in \mathcal{E}(K)$. We may perform the birational transformation $z = -x/y$, $w = -1/y$ so that (2.1) gives

$$\hat{\mathcal{E}} : w = g_3z^3 + g_2z^2w + g_1zw^2 + g_0w^3. \quad (2.2)$$

Thus one may obtain w as a power series in z ,

$$w = w(z) = g_3z^3 + g_3g_2z^5 + O(z^7) \in \mathbb{Z}[g_0, \dots, g_3][[z]]. \quad (2.3)$$

As the leading order is z^3 , we may write $1/x = w/z$ as a power series in z

$$\frac{1}{x} = \frac{1}{x}(z) = \frac{w(z)}{z} = g_3z^2 + g_3g_2z^4 + O(z^6) \in \mathbb{Z}[g_0, \dots, g_3][[z]]. \quad (2.4)$$

If (x_0, y_0) is another point on $\mathcal{E}$ then one may also express the x -coordinate of $(x, y) + (x_0, y_0)$ as a power series in z

$$\begin{aligned} x \text{ coordinate of } (x, y) + (x_0, y_0) &= x_0 + 2y_0z \\ &+ (3g_3x_0^2 + 2g_2x_0 + g_1)z^2 + (4g_3x_0y_0 + 2g_2y_0)z^3 \\ &+ (4g_3^2x_0^3 + 6g_2g_3x_0^2 + 2g_1g_3x_0 + 2g_2^2x_0 + g_0g_3 + g_3y_0^2 + g_1g_2)z^4 \\ &+ O(z^5) \in \mathbb{Z}[x_0, y_0, g_0, \dots, g_3][[z]]. \end{aligned} \quad (2.5)$$

The formal groups $\exp(t)$, $\log(t)$ associated to $\mathcal{E}$ could be computed to give

$$\log(t) = t + \frac{1}{3}g_2t^3 + \frac{1}{5}(g_2^2 + 2g_1g_3)t^5 + O(t^7) \in \mathbb{Q}[g_0, \dots, g_3][[t]], \quad (2.6)$$

$$\exp(t) = t - \frac{1}{3}g_2t^3 + \frac{1}{15}(2g_2^2 - 6g_1g_3)t^5 + O(t^7) \in \mathbb{Q}[g_0, \dots, g_3][[t]] \quad (2.7)$$

satisfying $\exp(s+t) = \mathcal{F}(\exp(s), \exp(t))$ and $\log(s) + \log(t) = \log(\mathcal{F}(s, t))$, where $\mathcal{F}$ is the formal group associated to $\mathcal{E}$ and s, t are two local parameters. (See [FW99, p. 9] for detailed calculations of (2.5), (2.6) and (2.7).)

2.2 Elliptic Curve Chabauty

We will now bound the number of K -rational points on $\mathcal{E}$ with x -coordinate $\mathbb{Q}$ -rational. To do so, we will follow the method in [FW99] and [Fly00]. Suppose that $K : \mathbb{Q}$ is a simple non-trivial extension, say $K = \mathbb{Q}(\alpha)$ for some α , and $[\mathbb{Q}(\alpha) : \mathbb{Q}] = d > 1$. Suppose the rank of $\mathcal{E}$ is r and so

$$\mathcal{E}(K) \cong \langle \mathcal{E}(K)_{\text{tors}}, P_1, \dots, P_r \rangle, \quad (2.8)$$

where $P_1, \dots, P_r$ are the generators of the torsion free part. When $r = 0$, there are only finitely many points to check, following Theorem 1.10.

Let p be an odd rational prime satisfying the following conditions:

1. $[\mathbb{Q}_p(\alpha) : \mathbb{Q}_p] = [\mathbb{Q}(\alpha) : \mathbb{Q}] = d$,
2. $|\alpha|_p = 1$,⁴
3. $|g_i|_p \leq 1$,
4. $\mathcal{E}$ has a good reduction at p ,
5. $\mathbb{F}_p(\tilde{\alpha})$ is the residue field of $\mathbb{Q}_p(\alpha)$,⁵
6. $\mathbb{Q}(\alpha)$ is unramified at p .

Thus let $\tilde{\mathcal{E}}$ be the reduction of $\mathcal{E}$ modulo p . Condition 3, 4 and 5 ensure that $\tilde{\mathcal{E}}$ is an elliptic curve over $\mathbb{F}_p(\tilde{\alpha})$.

Let $\tilde{P}_i$ be the mod p reduction of P_i . Since $\tilde{\mathcal{E}}(\mathbb{F}_p(\tilde{\alpha}))$ is finite, let m_i be the order of $\tilde{P}_i$. Define $Q_i = m_i P_i \in \mathcal{E}(\mathbb{Q}(\alpha))$, such points lie in kernel of the reduction map $\mathcal{E}(\mathbb{Q}(\alpha)) \rightarrow \tilde{\mathcal{E}}(\mathbb{F}_p(\tilde{\alpha}))$. Let $\mathcal{T}$ be the following finite set:

$$\left\{ S + \sum_{i=1}^r l_i P_i : S \in \mathcal{E}(K)_{\text{tors}}, \lfloor -m_i/2 \rfloor + 1 \leq l_i \leq \lfloor m_i/2 \rfloor \right\}. \quad (2.9)$$

⁴ $|\alpha|_p = \sqrt[d]{|\text{Norm}_{K:\mathbb{Q}}(\alpha)|_p}$.

⁵ $\tilde{\alpha}$ is the image of α under $\mathcal{O} \rightarrow \mathcal{O}/p\mathcal{O}$.

Thus, for every $L \in \mathcal{E}(K)$, there exists unique $T \in \mathcal{T}$ and $n_i \in \mathbb{Z}$ such that

$$L = T + \sum_{i=1}^r n_i Q_i. \quad (2.10)$$

Let z_i be the z -coordinate of each Q_i . The z -coordinate of $\sum_{i=1}^r n_i Q_i$ can be computed via $\exp(n_1 \log(z_1) + \cdots + n_r \log(z_r))$, using (2.6) and (2.7); this therefore gives a power series in $n_1, \dots, n_r$. Notice in either (2.6) or (2.7), the coefficient of t^j multiplied by $j!$ lie in $\mathbb{Z}[g_0, \dots, g_3]$ and $|j!|_p \geq p^{-(j-1)/p-1}$. Moreover, since Q_i is in kernel of reduction, $|z_i|_p \leq 1/p$. This shows that the z coordinate of $\sum_{i=1}^r n_i Q_i$ is a power series with coefficients in $\mathbb{Z}_p[\alpha]$, which has the property that the coefficient of $n_1^{v_1} \cdots n_r^{v_r}$ converges to 0 as $v_1 + \cdots + v_r \rightarrow \infty$.

If T in (2.10) is ∞ then the z -coordinate of L is the same as the z -coordinate of $\sum_{i=1}^r n_i Q_i$. By (2.4), one can therefore calculate the $1/(x\text{-coordinate of } L)$, as a power series in $n_1, \dots, n_r$. On the other hand, if $T \neq \infty$ then say $T = (x_0, y_0)$, one can therefore calculate the x coordinate of L via (2.5). We further assume that $\mathcal{T}$ does not contain any element in the kernel of reduction and so $|x_0|_p \leq 1, |y_0|_p \leq 1$.

With such Q_i chosen, this encourages the following well-defined function $\theta: \mathcal{T} \rightarrow \mathbb{Z}_p[\alpha][[n_1, \dots, n_r]]$:

$$\theta(T) = \begin{cases} 1/(x\text{-coordinate of } \sum_{i=1}^r n_i Q_i) & T = \infty \\ x\text{-coordinate of } T + \sum_{i=1}^r n_i Q_i & T \neq \infty. \end{cases} \quad (2.11)$$

Suppose we fix $T \in \mathcal{T}$. By condition 1, one may represent $\theta(T)$ over the basis $\{1, \alpha, \dots, \alpha^{d-1}\}$:

$$\theta(T) = \theta(T)^{(0)} + \theta(T)^{(1)}\alpha + \cdots + \theta(T)^{(d-1)}\alpha^{d-1}, \quad (2.12)$$

where $\theta(T)^{(i)} \in \mathbb{Z}_p[\alpha][[n_1, \dots, n_r]]$ and the coefficient of $n_1^{v_1} \cdots n_r^{v_r}$ converges to 0 as $v_1 + \cdots + v_r \rightarrow \infty$.

Suppose $(n_1, \dots, n_r) \in \mathbb{Z}^r$ is such that when $P = T + \sum_{i=1}^r n_i Q_i$, P has $\mathbb{Q}$ -rational x -coordinate. (2.12) shows that $\theta(T)^{(1)}(n_1, \dots, n_r) = \cdots = \theta(T)^{(d-1)}(n_1, \dots, n_r) = 0$. Thus the solutions to $\theta(T)^{(1)}(n_1, \dots, n_r) = \cdots = \theta(T)^{(d-1)}(n_1, \dots, n_r) = 0$ gives an upper bound of the number of points in the form of $T + \sum_{i=1}^r n_i Q_i$ with x -coordinate $\mathbb{Q}$ -rational. Now it remains to repeat the process with other $T \in \mathcal{T}$ to obtain a bound on the number of K -rational points on $\mathcal{E}$ with x -coordinate being $\mathbb{Q}$ -rational. One could also repeat this process with different p that satisfies the above conditions in order to produce a different bound.

When $r = 1$, the power series is in one variable and so one could use the following theorem to give a bound.

Theorem 2.1 (Strassman's Theorem). Suppose $\theta(X) = a_0 + a_1 X + a_2 X^2 + \cdots \in \mathbb{Z}_p[[X]]$ such that not all a_i are zero and $a_j \rightarrow 0$ as $j \rightarrow \infty$. If there exists N such that $|a_N|_p \geq |a_i|_p$ for all i and $|a_N|_p > |a_i|_p$ for all $i > N$ then there are at most N solutions in $\mathbb{Z}_p$ to $\theta(X)$.

In order to prove Strassman's Theorem, we will first prove a useful lemma.

Lemma 2.2 (Double summation). Let p be a rational prime. Suppose $a_{i,j} \in \mathbb{Q}$, $i, j = 0, 1, 2, \dots$, such that for every $\epsilon > 0$, there exists $\delta(\epsilon)$ such that whenever $\max(i, j) \geq \delta(\epsilon)$, $|a_{i,j}|_p < \epsilon$. Then, the series $\sum_i (\sum_j a_{i,j})$ and $\sum_j (\sum_i a_{i,j})$ both converge and are equal.

The Lemma and its proof are due to [Cas86], Lemma 4.1, with some elaboration from the author.

Proof. Let $\epsilon > 0$ be given and fix $i \in \mathbb{Z}^+$. $\sum_j a_{i,j}$ must converge since the tail is tending to 0.⁶ Moreover, whenever $i \geq \delta(\epsilon)$, $|\sum_j a_{i,j}|_p < \epsilon$ by the non-Archimedean property of the p -adic valuation. This shows that the tail of $\sum_i (\sum_j a_{i,j})$ again tends to 0 and thus $\sum_i (\sum_j a_{i,j})$ converges. Similarly, $\sum_j (\sum_i a_{i,j})$ converges as well. Thus it is left to show that they are equal.

Another application of the non-Archimedean property shows that

$$\left| \sum_{i=0}^{\delta(\epsilon)} \left(\sum_{j=0}^{\delta(\epsilon)} a_{i,j} \right) - \sum_i \left(\sum_j a_{i,j} \right) \right|_p \leq \max_{i,j > \delta(\epsilon)} |a_{i,j}|_p < \epsilon$$

and also

$$\left| \sum_{j=0}^{\delta(\epsilon)} \left(\sum_{i=0}^{\delta(\epsilon)} a_{i,j} \right) - \sum_j \left(\sum_i a_{i,j} \right) \right|_p \leq \max_{i,j > \delta(\epsilon)} |a_{i,j}|_p < \epsilon.$$

Notice that $\sum_{i=0}^{\delta(\epsilon)} (\sum_{j=0}^{\delta(\epsilon)} a_{i,j}) = \sum_{j=0}^{\delta(\epsilon)} (\sum_{i=0}^{\delta(\epsilon)} a_{i,j})$ as this is only a finite sum, an application of the triangle inequality shows that

$$\left| \sum_i \left(\sum_j a_{i,j} \right) - \sum_j \left(\sum_i a_{i,j} \right) \right|_p < 2\epsilon.$$

As $\epsilon > 0$ is arbitrary, we deduce that the double sums are equal. $\square$

Now we are in a position to prove Strassman's Theorem.

Strassman's Theorem and its proof are due to [Cas86], Theorem 4.1, with some elaboration from the author.

Proof. We will prove this by induction on N .

Suppose $N = 0$ and so $|a_0|_p > |a_i|_p$ for $i > 0$, we wish to show that $\theta(X)$ does not have any roots in $\mathbb{Z}_p$. Suppose $x \in \mathbb{Z}_p$ is a root then $|x|_p \leq 1$ and that $a_0 =$

⁶See C3.7 Lecture Notes, Theorem 2.12.

$-\sum_{i=1}^{\infty} a_i x^i$. We consider the p -adic valuation on both sides,

$$\begin{aligned}
\left| \sum_{i=1}^{\infty} a_i x^i \right|_p &\leq \max_{i \geq 1} |a_i x^i|_p && \text{by non-Archimedean property} \\
&\leq \max_{i \geq 1} |a_i|_p && \text{since } |x|_p \leq 1 \\
&< |a_0|_p && \text{by assumption.}
\end{aligned}$$

This therefore gives a contradiction and so indeed $\theta(X)$ does not have a root in $\mathbb{Z}_p$.

Next, assume $N > 0$, we will show that there are at most N solutions in $\mathbb{Z}_p$. If there are no solutions then the statement is trivially true. Suppose $x \in \mathbb{Z}_p$ is a solution and let $y \in \mathbb{Z}_p$. We have that

$$\begin{aligned}
\theta(y) - \theta(x) &= \sum_{i=1}^{\infty} a_i (y^i - x^i) \\
&= \sum_{i=1}^{\infty} a_i (y - x) (y^{i-1} + y^{i-2}x + \cdots + x^{i-1}) \\
&= \sum_{i=1}^{\infty} a_i (y - x) \sum_{j=0}^{i-1} (y^j x^{i-1-j}) \\
&= (y - x) \sum_{i=1}^{\infty} \sum_{j=0}^{i-1} a_i y^j x^{i-1-j}.
\end{aligned}$$

Since a_i tends to 0 as $i \rightarrow \infty$ and $|x|_p \leq 1, |y|_p \leq 1$, we can apply Lemma 2.2 so that

$$\sum_{i=1}^{\infty} \sum_{j=0}^{i-1} a_i y^j x^{i-1-j} = \sum_{n=0}^{\infty} \left(\sum_{m=0}^{\infty} a_{n+1+m} x^m \right) y^n.$$

Thus, $\theta(y) = (y - x) \sum_{n=0}^{\infty} \left(\sum_{m=0}^{\infty} a_{n+1+m} x^m \right) y^n$ and if y is a solution then either $y - x = 0$ or $\sum_{n=0}^{\infty} \left(\sum_{m=0}^{\infty} a_{n+1+m} x^m \right) y^n = 0$, this means that it is sufficient to show that there are at most $N - 1$ solutions to the power series $\sum_{n=0}^{\infty} \left(\sum_{m=0}^{\infty} a_{n+1+m} x^m \right) X^n$ to conclude this proof. For simplicity, let $h(X) = \sum_{n=0}^{\infty} \left(\sum_{m=0}^{\infty} a_{n+1+m} x^m \right) X^n$ and $g_n = \sum_{m=0}^{\infty} a_{n+1+m} x^m$.

In the case of $x = 0$, a_0 must be 0 and so we could consider $\theta(X)/X$ to complete the induction. Thus, we consider the case that $x \neq 0$, this means that g_n are not all zero. We shall proceed by checking the remaining conditions in the inductive hypothesis. It is also clear that $h(X) \in \mathbb{Z}_p[[X]]$ and so it remains to show that $g_j \rightarrow 0$ as $j \rightarrow \infty$, $|g_{N-1}|_p \geq |g_i|_p$ for all i and $|g_{N-1}|_p > |a_i|_p$ for all $i > N - 1$. Firstly, notice that

$$|g_j|_p = \left| \sum_{m=0}^{\infty} a_{j+1+m} x^m \right|_p \leq \max_{m \geq 0} |a_{j+1+m}|_p \rightarrow 0,$$

this means that $g_j \rightarrow 0$ as $j \rightarrow \infty$. Next, we claim $|g_{N-1}|_p = |\sum_{m=0}^{\infty} a_{N+m}x^m|_p = |a_N|_p$, this is merely due to $|a_N|_p > \max_{m \geq 1} |a_{N+m}|_p \geq \max_{m \geq 1} |a_{N+m}x^m|_p$. Thus, for all i , we have that

$$|g_i|_p = \left| \sum_{m=0}^{\infty} a_{i+1+m}x^m \right|_p \leq \max_{j \geq 0} |a_j|_p = |a_N|_p = |g_{N-1}|_p,$$

this therefore proves the second condition. Lastly, let $i > N - 1$, we have

$$|g_i|_p = \left| \sum_{m=0}^{\infty} a_{i+1+m}x^m \right|_p \leq \max_{j > 0} |a_{j+N}|_p < |a_N|_p = |g_{N-1}|_p.$$

After showing that $h(X)$ satisfies the inductive hypothesis at $N - 1$, one deduces that there are at most $N - 1$ solutions in $\mathbb{Z}_p$ to $h(X)$. Thus we have proved that there are at most N solutions in $\mathbb{Z}_p$ to $\theta(X)$ as required. $\square$

In the case that $r > 1$, provided that $r < d$, one could repeatedly apply the following theorem to obtain a power series in one variable and thus apply Strassman's Theorem. ([FW99, p. 13])

Theorem 2.3 (Weierstrass Preparation Theorem). Suppose $\theta(X) = a_0 + a_1X + a_2X^2 + \cdots \in \mathbb{Z}_p[[X]]$ is such that $|a_j|_p \rightarrow 0$ as $j \rightarrow \infty$. If there exists N such that $|a_N|_p \geq |a_i|_p$ for all i and $|a_N|_p > |a_i|_p$ for all $i > N$ then exists a polynomial $f(X) = f_0 + \cdots + f_NX^N \in \mathbb{Z}_p[X]$ and a power series $g(X) = 1 + g_1X + \cdots \in \mathbb{Z}_p[[X]]$ with $|g_j|_p \rightarrow 0$ as $j \rightarrow \infty$ such that $\theta(X) = f(X)g(X)$. Moreover, $|f_N|_p \geq |f_j|_p$ for all j and $|g_i|_p < 1$ for all $i \geq 1$.

Proof. See Chapter 6, Section 5 of [Cas86] for a proof. $\square$

Comment 2.4. In practice, it is difficult to produce a set of generators for the torsion free part; however, for this method, it is not imperative to do so. Suppose that the rank of the elliptic curve is 1, for simplicity, and one finds a torsion free element E . Same as the above, assume that $\tilde{E}$ has order m and let $Q = mE$. By Theorem 1.11, $\exists E'$ which generates the torsion free part and so $\exists W \in \mathbb{Z}$ such that $E = WE'$. If it could be shown that mE' also lies in the kernel of reduction and that $p \nmid W$ and $\text{hcf}(m, W) = 1$ then $L \in \mathcal{E}(K)$ can be written as $T + N \cdot Q$ where $T \in \{S + l \cdot E : S \in \mathcal{E}(K)_{\text{tors}}, \lfloor -m/2 \rfloor + 1 \leq l \leq \lfloor m/2 \rfloor\}$ and $N \in \mathbb{Z}_p$. Since Strassman's Theorem concerns about roots in $\mathbb{Z}_p$, the rest of the argument still flows. [See [Fly97, p. 12] and [CF96, p. 152] in the context of classical Chabauty for genus 2 curves.]

2.3 Worked Example

Example 2.5. The bound on the number of $\mathbb{Q}(i)$ -rational points with x -coordinate $\mathbb{Q}$ -rational on $\mathcal{E}_1 : Y^2 = F(X) = (X - 2)(X + 2)(X + 2i)$ is 5; in particular, there are precisely 5 $\mathbb{Q}(i)$ -rational points on $\mathcal{E}_1$ with x -coordinate $\mathbb{Q}$ -rational.

Proof. Recall in Example 1.18, $\mathcal{E}_1(\mathbb{Q}(i))_{\text{tors}} = \{\infty, A, B, C\}$ where $A = (2, 0)$, $B = (-2, 0)$ and $C = (-2i, 0)$. Denote $E = (2i, -4 + 4i)$, this is also a torsion free element. Using the notation in (2.1), $g_3 = 1, g_2 = 2i, g_1 = -4, g_0 = -8i$.

We consider $p = 3$, a finite check shows that such p satisfies conditions given in Section 2.2. In this case, let $\tilde{\mathcal{E}}_{1,3} : Y^2 = X^3 + 2iX^2 + 2X + i$, this is the reduction of $\mathcal{E}_1$ modulo 3. Magma shows that $\tilde{\mathcal{E}}_{1,3}(\mathbb{F}_3(i)) \cong \mathbb{Z}/2 \times \mathbb{Z}/6$ and $\tilde{E}$, the reduction of $E \bmod 3$, has order 6.

Since E is not necessarily a generator, we consider Comment 2.4. Suppose E' is a generator of the torsion free part of $\mathcal{E}_1(\mathbb{Q}(i))$, then $6E'$ lies in the kernel of reduction considering the group structure of $\tilde{\mathcal{E}}_{1,3}(\mathbb{F}_3(i))$. Suppose $E = WE'$ for some integer W , we wish to show that $3 \nmid W$. Suppose otherwise then $\frac{W}{3}$ is an integer and so $(\frac{W}{3}) \cdot 6E' = 2WE' = 2E$ is in the kernel of reduction, this gives a contradiction. Similarly, one could show that $2 \nmid W$. Thus for the purpose of this calculation, it is sufficient to treat E as a generator.

Let $\mathcal{T} = \{\infty, A, B, C, \pm E, \pm 2E, 3E, A \pm E, A \pm 2E, A + 3E, B \pm E, B \pm 2E, B + 3E, C \pm E, C \pm 2E, C + 3E\}$ and $Q = 6E$. Any $P \in \mathcal{E}_1(\mathbb{Q}(i))$ have the form $P = T + nQ$, $T \in \mathcal{T}, n \in \mathbb{Z}_3$. Unfortunately, one finds that $C + 3E$ lies in the kernel of reduction. For the purpose of this calculation let $R = C + 3E$ and $\mathcal{S} = \{\infty, A, B, C, \pm E, A \pm E, B \pm E, C \pm E\}$. Therefore, any $L \in \mathcal{E}_1(\mathbb{Q}(i))$ have the form $L = S + nR$, $S \in \mathcal{S}, n \in \mathbb{Z}_3$. A simple calculation using Magma shows that $\mathcal{S}$ contains no elements in the kernel of reduction and a calculation using SageMath shows the z -coordinate of R is $\frac{60}{287}(-1 + i)$ and that $\frac{60}{287}(-1 + i) = 6 + 21i \pmod{3^3}$. By (2.7) $\log(z\text{-coordinate of } R) = 24 + 3i \pmod{3^3}$, by (2.6),

$$z\text{-coordinate of } nR = (24 + 3i)n + (9 + 18i)n^3 \pmod{3^3}. \quad (2.13)$$

Assume P has $\mathbb{Q}$ -rational x -coordinate then $\tilde{P} = \tilde{S}$ and so $\tilde{P}$ has $\mathbb{F}_3$ -rational x -coordinate. Thus it is sufficient to only consider $S \in \mathcal{S}$ such that $\tilde{S}$ has $\mathbb{F}_3$ -rational x -coordinate. A finite search using SageMath shows that the only compatible elements are $\infty, A, B, C \pm E$.

We first consider ∞ . By (2.4), (2.13) gives

$$\theta(\infty) = O(n^2) = (24 + 3i)^2 n^2 \pmod{3^3}. \quad (2.14)$$

Breaking this up in the sense of (2.12), one requires $\theta(\infty)^{(1)}$ evaluated at n to be 0. Moreover, $\theta(\infty)^{(1)} = O(n^2) = 9n^2 \pmod{3^3}$, which has a double root at $n = 0$. By applying Strassman's Theorem to $\theta(\infty)^{(1)}/n^2$, one deduces that there are no solutions in $\mathbb{Z}_3$ for $\theta(\infty)^{(1)}/n^2$ and so one deduces the only possible n is 0. In other words, the only element in the form of nR with x -coordinate $\mathbb{Q}$ -rational is when $n = 0$. Indeed at $n = 0$, this just returns ∞ , which is perceived to have x -coordinate $\mathbb{Q}$ -rational.

Next, we consider $A = (2, 0)$. By (2.5) and (2.13)

$$\theta(A) = 2 + O(n^2) = 2 + (8 + 8i)(24 + 3i)^2 n^2 \pmod{3^3}. \quad (2.15)$$

Same as above $\theta(A)^{(1)} = O(n^2) = 18n^2 \pmod{3^3}$ and so $\theta(A)^{(1)}$ only has a root at $n = 0$. In other words, the only element in the form of $A + nR$ with x -coordinate $\mathbb{Q}$ -rational is when $n = 0$. A similar argument gives the same outcome for $B = (-2, 0)$.

Thus, it remains to consider $C \pm E = (0, \pm 2 \mp 2i)$. Start with $C + E$, one has

$$\begin{aligned} \theta(C + E) = O(n) = & 4(1 - i)((24 + 3i)n + (9 + 18i)n^3) \\ & - 4(24 + 3i)^2 n^2 \pmod{3^3}. \end{aligned} \quad (2.16)$$

Thus $\theta(C + E)^{(1)} = O(n) = 24n + 18n^2 + 9n^3 \pmod{3^3}$. Again, by applying Strassman's Theorem to $\theta(C + E)^{(1)}/n$, one finds that the only solution is $n = 0$. Notice that $C - E = -(C + E)$, this shows that the only solution for the case of $C - E$ is also only $n = 0$.

After showing that all relevant power series only have the solution at $n = 0$ and that all such $n = 0$ do indeed give a point with x -coordinate $\mathbb{Q}$ -rational, we conclude that the only points on $\mathcal{E}_1$ with x -coordinate $\mathbb{Q}$ -rational are $\infty, (2, 0), (-2, 0), (0, 2 - 2i)$ and $(0, -2 + 2i)$. $\square$

2.4 Some Extension on Elliptic Curve Chabauty

For the remaining of this chapter, we shall apply the same idea to finding $(x, y) \in \mathcal{E}(K)$ with $y \in \mathbb{Q}$. The following theory is original but is heavily influenced by Section 2.2.

Using the same notation and set up from the previous sections, $-\frac{1}{y}$ can be written as a power in z :

$$-\frac{1}{y} = -\frac{1}{y}(z) = w(z) = g_3 z^3 + g_3 g_2 z^5 + O(z^7) \in \mathbb{Z}[g_0, \dots, g_3][[z]]. \quad (2.17)$$

Let (x_0, y_0) be another point in $\mathcal{E}(K)$, one can also express y -coordinate of $(x, y) + (x_0, y_0)$ as a power series in z . To do so, first notice that the line connecting these two points is $Y = \ell X + (y_0 - \ell x_0)$ where $\ell = \frac{y - y_0}{x - x_0}$. One can then express ℓ as a Laurent expansion in z with coefficients in $\mathbb{Z}[x_0, y_0, g_1, \dots, g_3]$. Recall that, on an elliptic curve, three points sum to ∞ if and only if they are collinear. Thus y -coordinate of $(x, y) + (x_0, y_0)$ can be calculated as a Laurent expansion in z using $Y = \ell X + (y_0 - \ell x_0)$ and (2.5). Moreover, a simple computation shows that this is indeed a power series as the negative powers cancel and so one produces

$$\begin{aligned} y \text{ coordinate of } (x, y) + (x_0, y_0) = & y_0 + (3g_3 x_0^2 + 2g_2 x_0 + g_1)z \\ & + (6g_3 x_0 y_0 + 2g_2 y_0)z^2 + (2y_0^2 g_3 + x_0 g_3 (3g_3 x_0^2 + 2g_2 x_0 + g_1))z^3 \\ & + O(z^4) \in \mathbb{Z}[x_0, y_0, g_0, \dots, g_3][[z]]. \end{aligned} \quad (2.18)$$

Again, with such Q_i chosen and also assuming that $\mathcal{T}$ does not contain any element in kernel of reduction, define $\phi: \mathcal{T} \rightarrow \mathbb{Z}_p[\alpha][[n_1, \dots, n_r]]$:

$$\phi(T) = \begin{cases} -1/(y\text{-coordinate of } \sum_{i=1}^r n_i Q_i) & T = \infty \\ y\text{-coordinate of } T + \sum_{i=1}^r n_i Q_i & T \neq \infty. \end{cases} \quad (2.19)$$

Decompose $\phi(T)$ in the same sense as (2.12) to have $\phi(T) = \phi(T)^{(0)} + \phi(T)^{(1)}\alpha + \cdots + \phi(T)^{(d-1)}\alpha^{d-1}$. Thus, similar as before, we need to focus on $(n_1, \dots, n_r) \in \mathbb{Z}^r$ such that $\phi(T)^{(1)}(n_1, \dots, n_r) = \cdots = \phi(T)^{(d-1)}(n_1, \dots, n_r) = 0$. It is again merely an application of Strassman's Theorem and Weierstrass Preparation Theorem.

Example 2.6. The bound on the number of $\mathbb{Q}(i)$ -rational points with y -coordinate $\mathbb{Q}$ -rational on $\mathcal{E}_2 : Y^2 = F(X) = X^3 + 2X^2 + (-1 + 2i)X$ is 2; in particular, there are precisely 2 $\mathbb{Q}(i)$ -rational points on $\mathcal{E}_2$ with y -coordinate $\mathbb{Q}$ -rational.

Proof. In this case, $g_3 = 1$, $g_2 = 2$, $g_1 = (-1 + 2i)$, $g_0 = 0$. Magma shows that $\mathcal{E}_2(\mathbb{Q}(i))$ has rank 1 and $\mathcal{E}_2(\mathbb{Q}(i))_{\text{tors}} = \langle (0, 0) \rangle \cong C_2$. Let $P = (-2 - i, 3 - i)$, this is an element of infinite order. Moreover, since the discriminant is $-896 - 128i$, $p = 3$ again satisfies the conditions as it is a good reduction of $\mathcal{E}_2$. Let $\tilde{\mathcal{E}}_{2,3} : Y^2 = F(X) = X^3 + 2X^2 + (2 + 2i)X$, this is the reduction of $\mathcal{E}_2$ modulo 3. Furthermore, Magma also shows that $\tilde{\mathcal{E}}_{2,3}(\mathbb{F}_3(i)) \cong \mathbb{Z}/6$ and $\tilde{P}$ has order 6. Thus it is sufficient to treat P as a generator for the same reason as in Example 2.5.

Let $Q = 6P$ and $\mathcal{T} = \{\infty, \pm P, \pm 2P, 3P, (0, 0), (0, 0) \pm P, (0, 0) \pm 2P, (0, 0) + 3P\}$ such that any $L \in \mathcal{E}_2(\mathbb{Q}(i))$ can be written as $L = S + nQ$, $T \in \mathcal{T}$, $n \in \mathbb{Z}_p$. As $(0, 0) + 3P$ is in kernel of reduction again, we adjust analogously to Example 2.5. Let $R = (0, 0) + 3P$ and $\mathcal{S} = \{\infty, \pm P, (0, 0), (0, 0) \pm P\}$. Therefore, any $P \in \mathcal{E}_1(\mathbb{Q}(i))$ have the form $P = S + nR$, $S \in \mathcal{S}$, $n \in \mathbb{Z}_3$. Moreover, $\mathcal{S}$ contains no elements in kernel of reduction, by checking with Magma. Lastly, SageMath shows that the z -coordinate of $R = -\frac{1}{50}(51 - 93i) = 33 + 240i \pmod{3^5}$. By (2.6) and (2.7),

$$z\text{-coordinate of } nR = (69 + 42i)n + (126 + 117i)n^3 + (81 + 81i)n^5 \pmod{3^5}. \quad (2.20)$$

Assume $L = S + nR$ has $\mathbb{Q}$ -rational y -coordinate then $\tilde{S}$ has $\mathbb{F}_p$ -rational y -coordinate. A finite search shows that the only compatible elements are $\infty, (0, 0)$.

Start with ∞ . By (2.17)

$$\phi(\infty) = O(n^3) = (54 + 189i)n^3 \pmod{3^5}. \quad (2.21)$$

Decompose $\phi(\infty) = \phi(\infty)^{(0)} + \phi(\infty)^{(1)}i$, one obtains $\phi(\infty)^{(1)} = O(n^3) = 189n^3 \pmod{3^5}$. Thus, by applying Strassman's Theorem on $\phi(\infty)^{(1)}/n^3$, we see that the only root is $n = 0$. This means that the only element in the form of nR with y -coordinate $\mathbb{Q}$ -rational is ∞ .

We repeat the computation with $(0, 0)$. In fact, it is sufficient to just work over $\pmod{3^4}$. In this case, z -coordinate of $nR = (69 + 42i)n + (45 + 36i)n^3 \pmod{3^4}$. Thus by (2.18)

$$\begin{aligned} \phi(A) = O(n) &= (-1 + 2i)((69 + 42i)n + (45 + 36i)n^3) \\ &= (90 + 96i)n + (126 + 54i)n^3 \pmod{3^4}. \end{aligned} \quad (2.22)$$

Decompose as above, $\phi(A)^{(1)} = O(n) = 96n + 54n^3 \pmod{3^4}$. Since $|96|_3 = 3^{-1}$ and $|54|_3 = 3^{-3}$, we could apply Strassman's Theorem to $\phi(A)^{(1)}/n$ to deduce that the

only solution to $\phi(A)^{(1)}$ is $n = 0$. This means that the only element in the form of $(0, 0) + nR$ with y -coordinate $\mathbb{Q}$ -rational is $(0, 0)$.

After considering all cases, we deduce that the only $\mathbb{Q}(i)$ -rational points on $\mathcal{E}_2$ with y -coordinate $\mathbb{Q}$ -rational are ∞ and $(0, 0)$.

□

3 Group Structure on Genus 2 Curves

We will now shift our attention to curves of genus 2 over K . In particular, we focus on hyperelliptic curves of the following form

$$\mathcal{G} : Y^2 = G(X) = g_6X^6 + g_5X^5 + g_4X^4 + g_3X^3 + g_2X^2 + g_1X + g_0.$$

where $G(X) \in K[X]$ with $\Delta_G \neq 0$ and $g_6 \neq 0$ or $g_5 \neq 0$. Further, we assume $\text{char}K \neq 2, 3$ or 5 .

3.1 Group Structure of a Genus 2 Curve

Let $\mathcal{G} : Y^2 = G(X)$ be a curve of genus 2 over K . Classically, the commutative group on $\mathcal{G}$ is understood via the Picard group, denoted by $\text{Pic}^0(\mathcal{G})$. This is the group of degree 0 divisors on $\mathcal{G}$ modulo linear equivalence. Moreover, [Sil94, p. 197–199] shows that $\text{Pic}^0(\mathcal{G})$ could be identified with the *Jacobian*, where the latter is denoted by $\mathcal{J}(\mathcal{G})$. Additionally, let $\mathcal{J}(K)$ denote the subgroup where the elements are invariant under Galois transformations, this is called the **Mordell-Weil group**. For practical reasons, we shall provide an explicit construction of $\mathcal{J}(\mathcal{G})$, following [Fly93, p. 6].

Suppose $P = (x, y)$ is a point on $\mathcal{G}$, let $\bar{P} = (x, -y)$, then clearly $\bar{P}$ is also a point on $\mathcal{G}$. $\mathcal{J}(\mathcal{G})$ consists of *unordered* pairs of points on $\mathcal{G}$, allowing points at infinity. If $\deg G(X) = 5$ then there is only one point at infinity, denoted by ∞ ; if $\deg G(X) = 6$ then there are two points at infinity, denoted by $\infty^\pm$. The identity element of this group is the canonical equivalence class, $\mathcal{O}$, and can be identified via any unordered pair of the form $\{P, \bar{P}\}$. Furthermore, the group laws are governed by $-(P, Q) = (\bar{P}, \bar{Q})$ and $(P_1, Q_1) + (P_2, Q_2) + (P_3, Q_3) = \mathcal{O}$ if there exists $M(X) \in K[X]$ with degree at most 3 such that $Y = M(X)$ meets $\mathcal{G}$ at the above 6 points.

With such description, the Mordell-Weil group consists of pairs of points where both are K rational or both are defined over a quadratic extension over K and conjugate over K .⁷

Definition 3.1. A **Weierstrass point** on $\mathcal{G} : Y^2 = G(X)$ is a point P such that $P = \bar{P}$.

⁷See Appendix 6.2 on when points at infinity lie in $\mathcal{J}(\mathbb{Q})$.

Example 3.2. In order to determine $\mathcal{J}(K)[2]$, it is enough to determine the roots of $G(X)$, given the definition of $-(P, Q)$. It is, therefore, sufficient to understand the Weierstrass points on $\mathcal{G}$. We shall give two examples of this.

1. Suppose $\mathcal{G} : Y^2 = G(X) = (X - x_1)(X - x_2)(X - x_3)(X - x_4)(X - x_5)$ over K . There are 6 K -rational Weierstrass points, $(x_1, 0), (x_2, 0), (x_3, 0), (x_4, 0), (x_5, 0)$ and ∞ . Thus, there are 15 distinct elements of $\mathcal{J}(K)$ of order 2, since $\{(x_j, 0), (x_j, 0)\}$ and $\{\infty, \infty\}$ are all identified as $\mathcal{O}$ for $j = 1, 2, 3, 4, 5$. Together with $\mathcal{O}$, $\mathcal{J}(K)[2]$ is a Boolean group consisting of 16 elements.
2. Suppose $\mathcal{G} : Y^2 = G(X) = G_1(X)G_2(X)G_3(X)$ over K where each $G_i(X)$ is a quadratic. Suppose further that $G_1(X)$ and $G_2(X)$ are irreducible while $G_3(X)$ factors into linear factors. Let a_i, b_i be the roots to $G_i(X)$. Since $G_1(X)$ is irreducible, a_1, b_1 both only lie in $K[X]/\langle G_1(X) \rangle$, a quadratic extension of K , and are conjugate of each other. Similarly for a_2, b_2 . On the other hand a_3, b_3 are the only K -rational Weierstrass points on $\mathcal{G}$. Thus $\mathcal{J}(K)[2] = \{\mathcal{O}, \{(a_1, 0), (b_1, 0)\}, \{(a_2, 0), (b_2, 0)\}, \{(a_3, 0), (b_3, 0)\}\} \cong C_2 \times C_2$.

Perhaps quite surprisingly, the following theorems for curves of genus 2 are completely analogous to the ones for elliptic curves.

Theorem 3.3 ([CF96, p. 78]). Suppose $\mathcal{G} : Y^2 = G(X)$ is a curve of genus 2 over $\mathbb{Q}$. Let p be a rational prime such that $p \nmid 2\Delta_G$, then $\tilde{\mathcal{G}}$, the reduction of $\mathcal{G}$ modulo p , is also a curve of genus 2, now over $\mathbb{F}_p$. Moreover, let $\mathcal{J}(\mathbb{Q}), \tilde{\mathcal{J}}(\mathbb{F}_p)$ be the Mordell-Weil group of $\mathcal{G}, \tilde{\mathcal{G}}$ respectively, then $\mathcal{J}(\mathbb{Q})_{\text{tors}} \hookrightarrow \tilde{\mathcal{J}}(\mathbb{F}_p)$.

Theorem 3.4 (Mordell-Weil Theorem, [Fly00, p. 5]). Suppose $\mathcal{G}$ is a curve of genus 2 over K , where K is a number field, then $\mathcal{J}(K)$ is finitely generated.

Corollary 3.5. $\mathcal{J}(K) \cong \mathcal{J}(K)_{\text{tors}} \times \mathbb{Z}^r; r \in \mathbb{N}_0$. r is known as the **rank** of the Jacobian.

Comment 3.6. Analogously, $\mathcal{J}(K)/2\mathcal{J}(K) \cong \mathcal{J}(K)[2] \times [C_2]^{\times r}$ and $\mathcal{J}(K)[2]$ can be found using the ideas in Example 3.2. This isomorphism will again aid the rank calculations.

3.2 Finding the Rank of Jacobian

To compute the rank, we will investigate two methods, one via complete 2 descent and the other via Richelot Isogeny; both techniques will be accompanied by considering commutative diagrams similar to [Figure 1]. We will examine these in detail, while following [CF96, Chapter 11].

3.2.1 Rank computation via complete 2 descent

In this section, we assume that the curve of genus 2 is defined over $\mathbb{Q}$ and has rational roots so that it can be birationally transformed into the following form:

$$\mathcal{G} : Y^2 = G(X) = (X - x_1)(X - x_2)(X - x_3)(X - x_4)(X - x_5),$$

where $x_i \in \mathbb{Z}$.

Completely analogous to Section 1.2.2, we consider the following map:

$$\begin{aligned} \mu : \mathcal{J}(\mathbb{Q})/2\mathcal{J}(\mathbb{Q}) &\longrightarrow [\mathbb{Q}^*/(\mathbb{Q}^*)^2]^{\times 4} \\ \{P, Q\} &\mapsto [x - x_1, x - x_2, x - x_3, x - x_4] \times [u - x_1, u - x_2, u - x_3, u - x_4] \end{aligned}$$

where P has coordinates (x, y) and Q has coordinates (u, v) .

Comment 3.7. We interpret $[x - x_1, x - x_2, x - x_3, x - x_4]$ analogously to Lemma 1.12 when P is a $\mathbb{Q}$ -rational Weierstrass point or ∞ , vice versa for Q .

Let $\mathcal{S} := \{p : p \text{ prime}, p | 2\Delta_G\}$, this is the set of bad primes of $\mathcal{G}$. Since $\mathcal{S}$ is finite, $\mathcal{S} = \{p_1, \dots, p_m\}$ and let $\mathbb{Q}(\mathcal{S}) = \{\pm p_1^{r_1} \cdots p_m^{r_m} : r_i = 0, 1\}$. The following crucial result anchors the method of complete 2 descent.

Lemma 3.8. μ is an injective group homomorphism and $\text{Im}\mu \leq [\mathbb{Q}(\mathcal{S})]^{\times 4}$.

Proof. See [CF96, Chapter 11]. □

The image of μ is again bounded using a local analysis as we consider the commutative diagram in [Figure 2]; the maps are defined analogously to the ones in Section 1.2.2. Again, $j_p^{-1}(\text{Im}\mu_p)$ will provide an upper bound for $\text{Im}\mu$ for every rational prime p .

$$\begin{array}{ccc} \mathcal{J}(\mathbb{Q})/2\mathcal{J}(\mathbb{Q}) & \xrightarrow{\mu} & [\mathbb{Q}(\mathcal{S})]^{\times 4} \\ i_p \downarrow & & \downarrow j_p \\ \mathcal{J}(\mathbb{Q}_p)/2\mathcal{J}(\mathbb{Q}_p) & \xrightarrow{\mu_p} & [\mathbb{Q}_p^*/(\mathbb{Q}_p^*)^2]^{\times 4} \end{array}$$

Figure 2: Complete 2 descent - Curves of Genus 2

The following Lemma is merely an extension of Lemma 1.17 to the setting of curves of genus 2. See [CF96, Chapter 7] for more detail.

Lemma 3.9. Let $\mathcal{G}$ be a curve of genus 2 defined over $\mathbb{Q}$ and let $\mathcal{J}(\mathbb{Q})$ be the Mordell-Weil group. Suppose p is a rational prime then

$$\#\mathcal{J}(\mathbb{Q}_p)/2\mathcal{J}(\mathbb{Q}_p) = \begin{cases} \#\mathcal{J}(\mathbb{Q}_p)[2]/4 & p = \infty \\ \#\mathcal{J}(\mathbb{Q}_p)[2] \times 4 & p = 2 \\ \#\mathcal{J}(\mathbb{Q}_p)[2] & p \neq 2, \infty. \end{cases}$$

Since we will have to work with $\mathbb{Q}_p^*/(\mathbb{Q}_p^*)^2$ during the computation, it is worth noting the following result.

Lemma 3.10. Let p be a rational prime. If $p \neq 2$ then $\mathbb{Q}_p^*/(\mathbb{Q}_p^*)^2$ has a set of representatives of the form $\{1, u, p, up\}$ where $(\frac{u}{p}) = -1$. If $p = 2$ then $\mathbb{Q}_2^*/(\mathbb{Q}_2^*)^2$ has a set of representatives $\{\pm 1, \pm 2, \pm 5, \pm 10\}$.

Proof. See Theorem 3 and 4 of [Ser12]. $\square$

Comment 3.11. When appropriate, we take a set of representatives of $\mathbb{Q}_2^*/(\mathbb{Q}_2^*)^2$ to be $\{\pm 1, \pm 2, \pm 3, \pm 6\}$. This is because by Hensel's Lemma, $-\frac{3}{5}$ is a square in $\mathbb{Q}_2^*$.

The following original example is inspired by Example 11.3.1 in [CF96] and uses all the tools described above.

Example 3.12. Let $\mathcal{G}_1 : Y^2 = \hat{G}_1(X) = X(X-1)(X+1)(X+2)(X+4)$, this has rank 1 over $\mathbb{Q}$ and $\mathcal{J}(\mathbb{Q})_{\text{tors}} = \mathcal{J}(\mathbb{Q})[2]$.

Proof. By Example 3.2, $\mathcal{J}(\mathbb{Q})_{\text{tors}}$ contains a 2-torsion subgroup of order 16. The discriminant in this case is $2^{10}3^45^2$ and so the bad primes are 2, 3 and 5. Magma shows that $\#\tilde{\mathcal{J}}(\mathbb{F}_7) = 48$ and $\#\tilde{\mathcal{J}}(\mathbb{F}_{11}) = 176$, thus $\mathcal{J}(\mathbb{Q})_{\text{tors}} = \mathcal{J}(\mathbb{Q})[2]$, by Theorem 3.3. Moreover, $\{(0, 0), \infty\}$, $\{(1, 0), \infty\}$, $\{(-1, 0), \infty\}$ and $\{(-2, 0), \infty\}$ generate $\mathcal{J}(\mathbb{Q})_{\text{tors}}$.

Moreover, $\mathbb{Q}(\mathcal{S}) = \{\pm 1, \pm 2, \pm 3, \pm 5, \pm 6, \pm 10, \pm 15, \pm 30\} = \langle -1, 2, 3, 5 \rangle$. Thus by Lemma 3.8,

$$\begin{aligned} \text{Im}\mu \leq [\mathbb{Q}(\mathcal{S})]^{\times 4} = & \langle [-1, 1, 1, 1], [2, 1, 1, 1], [3, 1, 1, 1], [5, 1, 1, 1], \\ & [1, -1, 1, 1], [1, 2, 1, 1], [1, 3, 1, 1], [1, 5, 1, 1], \\ & [1, 1, -1, 1], [1, 1, 2, 1], [1, 1, 3, 1], [1, 1, 5, 1], \\ & [1, 1, 1, -1], [1, 1, 1, 2], [1, 1, 1, 3], [1, 1, 1, 5] \rangle. \end{aligned} \quad (3.1)$$

We have, under μ , that

$$\begin{aligned} \{(0, 0), \infty\} &\mapsto [-2, -1, 1, 2], \\ \{(1, 0), \infty\} &\mapsto [1, 30, 2, 3], \\ \{(-1, 0), \infty\} &\mapsto [-1, -2, 6, 1], \\ \{(-2, 0), \infty\} &\mapsto [-2, -3, -1, -3], \\ \{(2, 12), \infty\} &\mapsto [2, 1, 3, 1], \end{aligned}$$

where $\{(2, 12), \infty\} \in \mathcal{J}(\mathbb{Q})$ has infinite order.

Let $H = \langle [-2, -1, 1, 2], [1, 30, 2, 3], [-1, -2, 6, 1], [-2, -3, -1, -3], [2, 1, 3, 1] \rangle$. In order to see that the curve has rank 1, it is therefore sufficient to show that $H = \text{Im}\mu$. To do so, it is also enough to show

$$H = \bigcap_p j_p^{-1}(\text{Im}(\mu_p)). \quad (3.2)$$

We begin by noticing that $\#\mathcal{J}(\mathbb{Q}_p)[2] = 16$ for every p as all the roots to $\hat{G}_1$ are $\mathbb{Q}$ -rational.

Firstly, we consider $p = \infty$ and so $\mathbb{Q}_\infty = \mathbb{R}$. A chosen set of representative of $\mathbb{R}^*/(\mathbb{R}^*)^2$ is $\{\pm 1\}$. Under μ_∞ we have

$$\begin{aligned}\{(0, 0), \infty\} &\mapsto [-1, -1, 1, 1], \\ \{(1, 0), \infty\} &\mapsto [1, 1, 1, 1], \\ \{(-1, 0), \infty\} &\mapsto [-1, -1, 1, 1], \\ \{(-2, 0), \infty\} &\mapsto [-1, -1, -1, -1], \\ \{(2, 12), \infty\} &\mapsto [1, 1, 1, 1].\end{aligned}$$

By Lemma 3.9, $\#\mathcal{J}(\mathbb{Q}_\infty)/2\mathcal{J}(\mathbb{Q}_\infty) = 4$ and so $\mathcal{J}(\mathbb{Q}_\infty)/2\mathcal{J}(\mathbb{Q}_\infty)$ is a Boolean group with 2 generators. Because μ_∞ is injective, we choose the two generators of $\mathcal{J}(\mathbb{Q}_\infty)/2\mathcal{J}(\mathbb{Q}_\infty)$ to be $\{(0, 0), \infty\}$ and $\{(-2, 0), \infty\}$ and so $\text{Im}(\mu_\infty) = \langle [-1, -1, 1, 1], [-1, -1, -1, -1] \rangle$. Notice that $j_\infty^{-1}(\text{Im}\mu_\infty) = \langle \ker j_\infty, H \rangle$ and the elements in $\mathbb{Q}(\mathcal{S})$ that reduce to the identity in $\mathbb{R}^*/(\mathbb{R}^*)^2$ is a subgroup generated by $\{2, 3, 5\}$. One obtains

$$\begin{aligned}\text{Im}(\mu) \leq j_\infty^{-1}(\text{Im}\mu_\infty) &= \langle \ker j_\infty, H \rangle = \langle [2, 1, 1, 1], [3, 1, 1, 1], [5, 1, 1, 1], [1, 2, 1, 1], \\ &\quad [1, 3, 1, 1], [1, 5, 1, 1], [1, 1, 2, 1], [1, 1, 3, 1], \\ &\quad [1, 1, 5, 1], [1, 1, 1, 2], [1, 1, 1, 3], [1, 1, 1, 5], \\ &\quad [-1, -1, 1, 1], [-1, -1, -1, -1] \rangle.\end{aligned}\tag{3.3}$$

For $p = 3$, a chosen set of representative of $\mathbb{Q}_3^*/(\mathbb{Q}_3^*)^2$ is $\{\pm 1, \pm 3\}$, by Lemma 3.10. By Lemma 3.9, we have that $\#\mathcal{J}(\mathbb{Q}_3)/2\mathcal{J}(\mathbb{Q}_3) = 16$ and so it is sufficient to produce 4 generators. Using that -2 is a square in $\mathbb{Q}_3^*$, we have the following images under the map μ_3

$$\begin{aligned}\{(0, 0), \infty\} &\mapsto [1, -1, 1, -1], \\ \{(1, 0), \infty\} &\mapsto [1, 3, -1, 3], \\ \{(-1, 0), \infty\} &\mapsto [-1, 1, -3, 1], \\ \{(-2, 0), \infty\} &\mapsto [1, -3, -1, -3], \\ \{(2, 12), \infty\} &\mapsto [-1, 1, 3, 1].\end{aligned}$$

Since $\{[1, -1, 1, -1], [1, 3, -1, 3], [-1, 1, -3, 1], [-1, 1, 3, 1]\}$ form a linearly independent set, by considering the last coordinate if necessary, $\{\{(0, 0), \infty\}, \{(1, 0), \infty\}, \{(-1, 0), \infty\}, \{(2, 12), \infty\}\}$ form a set of generators for $\mathcal{J}(\mathbb{Q}_3)/2\mathcal{J}(\mathbb{Q}_3)$. So we have

$$\text{Im}\mu_3 = \langle [1, -1, 1, -1], [1, 3, -1, 3], [-1, 1, -3, 1], [-1, 1, 3, 1] \rangle.\tag{3.4}$$

The elements in $\mathbb{Q}(\mathcal{S})$ that reduce to the identity in $\mathbb{Q}_3^*/(\mathbb{Q}_3^*)^2$ is a subgroup generated

by $\{-2, -5\}$ and so one obtains

$$\begin{aligned} \text{Im}\mu \leq j_3^{-1}(\text{Im}\mu_3) = \langle \ker j_3, H \rangle = \langle [-2, 1, 1, 1], [-5, 1, 1, 1], [1, -2, 1, 1], [1, -5, 1, 1], \\ [1, 1, -2, 1], [1, 1, -5, 1], [1, 1, 1, -2], [1, 1, 1, -5], \\ [1, -1, 1, -1], [1, 3, -1, 3], [-1, 1, -3, 1], [-1, 1, 3, 1] \rangle. \end{aligned} \quad (3.5)$$

For $p = 5$, we choose a set of representatives to be $\{1, 2, 5, 10\}$.

Under μ_5 , we have

$$\begin{aligned} \{(0, 0), \infty\} &\mapsto [2, 1, 1, 2], \\ \{(1, 0), \infty\} &\mapsto [1, 5, 2, 2], \\ \{(-1, 0), \infty\} &\mapsto [1, 2, 1, 1], \\ \{(-2, 0), \infty\} &\mapsto [2, 2, 1, 2], \\ \{(2, 12), \infty\} &\mapsto [2, 1, 2, 1]. \end{aligned}$$

One could notice that $[2, 1, 1, 2], [1, 5, 2, 2], [1, 2, 1, 1], [2, 1, 2, 1]$ are four linearly independent elements and that the elements in $\mathbb{Q}(\mathcal{S})$ that reduce to the identity in $\mathbb{Q}_5^*/(\mathbb{Q}_5^*)^2$ is a subgroup generated by $\{-1, 6\}$ and so we have

$$\begin{aligned} \text{Im}\mu \leq j_5^{-1}(\text{Im}\mu_5) = \langle \ker j_5, H \rangle = \langle [-1, 1, 1, 1], [6, 1, 1, 1], [1, -1, 1, 1], [1, 6, 1, 1], \\ [1, 1, -1, 1], [1, 1, 6, 1], [1, 1, 1, -1], [1, 1, 1, 6], \\ [2, 1, 1, 2], [1, 5, 2, 2], [1, 2, 1, 1], [2, 1, 2, 1] \rangle. \end{aligned} \quad (3.6)$$

Lastly, we repeat for $p = 2$. As in Comment 3.11, a chosen set of representatives is $\{\pm 1, \pm 2, \pm 3, \pm 6\}$. Under μ_2 , we have

$$\begin{aligned} \{(0, 0), \infty\} &\mapsto [-2, -1, 1, 2], \\ \{(1, 0), \infty\} &\mapsto [1, -2, 2, 3], \\ \{(-1, 0), \infty\} &\mapsto [-1, -2, 6, 1], \\ \{(-2, 0), \infty\} &\mapsto [-2, -3, -1, -3], \\ \{(2, 12), \infty\} &\mapsto [2, 1, 3, 1]. \end{aligned}$$

By considering the last coordinate if necessary, we see that these 5 elements are linearly independent. In the light of Lemma 3.9, there is precisely one generator missing. A short search reveals that $\{(8, \alpha), \infty\} \in \mathcal{J}(\mathbb{Q}_2)$ and is mapped to $[2, -1, 1, -6]$. Such α exists since $\frac{\hat{G}_1(8)}{64} = 945 \equiv 1 \pmod{8}$ and so $\hat{G}_1(8)$ is a square in $\mathbb{Q}_2$. One notice that $[2, -1, 1, -6]$ is independent to the above 5 elements.

The elements in $\mathbb{Q}(\mathcal{S})$ that reduce to the identity in $\mathbb{Q}_2^*/(\mathbb{Q}_2^*)^2$ is a subgroup generated by $\{-15\}$ and so one achieves

$$\begin{aligned} \text{Im}\mu \leq j_2^{-1}(\text{Im}\mu_2) = \langle \ker j_2, H \rangle = \langle [-15, 1, 1, 1], [1, -15, 1, 1], \\ [1, 1, -15, 1], [1, 1, 1, -15], [-2, -1, 1, 2], [1, -2, 2, 3], \\ [-1, -2, 6, 1], [-2, -3, -1, -3], [2, 1, 3, 1], [2, -1, 1, -6] \rangle. \end{aligned} \quad (3.7)$$

In fact, these primes are sufficient as

$$H = \bigcap_{p=2,3,5,\infty} j_p^{-1}(\text{Im}(\mu_p)). \quad (3.8)$$

While SageMath (see Appendix of [Lan]) computes such result in a timely manner, we will obtain (3.8) by hand. (Inspired by [CF96, p. 124].)

Recall that $H = \langle [-2, -1, 1, 2], [1, 30, 2, 3], [-1, -2, 6, 1], [-2, -3, -1, -3], [2, 1, 3, 1] \rangle$ and so the second coordinates of the elements in H span all of $\mathbb{Q}(\mathcal{S})$. Moreover, the only elements in H with the second coordinate being 1 are $[2, 1, 3, 1]$ and $[1, 1, 1, 1]$. The aim here is to show that the only elements in $\bigcap_p j_p^{-1}(\text{Im}(\mu_p))$ with second coordinate being 1 are $[2, 1, 3, 1]$ and $[1, 1, 1, 1]$ and thus (3.8) is reached.

Suppose $[a, 1, c, d] \in \bigcap_p j_p^{-1}(\text{Im}(\mu_p))$. With $p = \infty$, we quickly deduce that a is positive. Using $p = 3$ and $p = 5$, one deduces that $a = 1$ or $a = 2$ by considering all possible combinations. When $a = 1$, a finite computation shows that $c = d = 1$ by considering the third and fourth coordinate using $p = 2$, $p = 3$ and $p = 5$. Now we consider $a = 2$. It's again a finite computation to realise the only compatible element in $j_2^{-1}(\text{Im}(\mu_2))$ with $a = 2$ is $[2, 1, 3, 1]$.

We therefore conclude that (3.8) is indeed true and so $\mathcal{J}(\mathbb{Q})/2\mathcal{J}(\mathbb{Q}) \cong [C_2]^{\times 5}$. Recall that $\mathcal{J}(\mathbb{Q})/2\mathcal{J}(\mathbb{Q}) \cong \mathcal{J}(Q)[2] \times [C_2]^{\times r}$ and $\mathcal{J}(Q)[2] \cong [C_2]^{\times 4}$. One concludes that the rank of $\mathcal{G}$ is 1 over $\mathbb{Q}$. $\square$

Comment 3.13. It is required that $G(X) \in \mathbb{Q}[X]$ has odd degree, in this case 5; when degree of $G(X)$ is even then the map μ is not necessarily injective (see Lemma 5.1 of [Sto01]). However, it is not imperative that all roots of $G(X)$ should lie in $\mathbb{Q}$. Such principles also hold in the context of elliptic curves (Section 1.2.2). Suppose $G(X) = G_1(X) \cdots G_k(X)$ where each $G_i(X)$ is irreducible, let θ_i be a root of $G_i(X)$. Then there exists an embedding $\mu' : \mathcal{J}(\mathbb{Q})/2\mathcal{J}(\mathbb{Q}) \rightarrow \mathbb{Q}(\theta_1)^*/(\mathbb{Q}(\theta_1)^*)^2 \times \cdots \times \mathbb{Q}(\theta_k)^*/(\mathbb{Q}(\theta_k)^*)^2$. See Section 6 of [AJF] for a worked example.

3.2.2 Rank computation via Richelot Isogeny

The idea of Richelot Isogeny is analogous to 2-isogeny encountered in Chapter 1. Suppose the curve of genus 2 over $\mathbb{Q}$ is now of the form of

$$\mathcal{G} : Y^2 = G(X) = G_1(X)G_2(X)G_3(X),$$

where each $G_i(X)$ is a quadratic. Suppose further that $G_i(X) = a_iX^2 + b_iX + c_i$ for $i = 1, 2, 3$.

For two polynomials g, h , define $(g, h) = g'h - gh'$. With such notation, define $L_1(X) = (G_2(X), G_3(X))$, $L_2(X) = (G_3(X), G_1(X))$ and $L_3(X) = (G_1(X), G_2(X))$. Let $b_{ij} = \text{Resultant}(G_i(X), G_j(X))$ and $\hat{b}_{ij} = \text{Resultant}(L_i(X), L_j(X))$. Lastly, define Δ as

$$\Delta = \begin{vmatrix} c_1 & b_1 & a_1 \\ c_2 & b_2 & a_2 \\ c_3 & b_3 & a_3 \end{vmatrix}.$$

We assume $\mathcal{G}$ is such that $\Delta \neq 0$ to avoid degenerate cases in the following lemmas

Lemma 3.14. Let $\hat{\mathcal{G}} : \Delta Y^2 = G(X) = L_1(X)L_2(X)L_3(X)$. Denote $\mathcal{J}(\hat{\mathcal{G}})$ to be the Jacobian of $\hat{\mathcal{G}}$. Then $\exists \phi : \mathcal{J}(\mathcal{G}) \rightarrow \mathcal{J}(\hat{\mathcal{G}})$ and $\exists \hat{\phi} : \mathcal{J}(\hat{\mathcal{G}}) \rightarrow \mathcal{J}(\mathcal{G})$ satisfying the following properties

1. ϕ and $\hat{\phi}$ are 4 isogenies.
2. $\ker \phi \cong [C_2]^{\times 2}$, $\ker \hat{\phi} \cong [C_2]^{\times 2}$.
3. $\hat{\phi} \circ \phi = [2]_{\mathcal{J}(\mathcal{G})}$, $\phi \circ \hat{\phi} = [2]_{\mathcal{J}(\hat{\mathcal{G}})}$.

Furthermore, we have the following exact sequence

$$0 \longrightarrow \ker \hat{\phi} \longrightarrow \mathcal{J}(\hat{\mathbb{Q}})/\phi(\mathcal{J}(\mathbb{Q})) \longrightarrow \mathcal{J}(\mathbb{Q})/2\mathcal{J}(\mathbb{Q}) \longrightarrow \mathcal{J}(\mathbb{Q})/\hat{\phi}(\mathcal{J}(\hat{\mathbb{Q}})) \longrightarrow 0.$$

We say $\hat{\mathcal{G}}$ and $\mathcal{G}$ are isogenous or $\hat{\mathcal{G}}$ is a curve isogenous to $\mathcal{G}$.

Proof. For a detailed construction, see Chapter 2 of [Fly94]. □

With Lemma 3.14, it is enough to compute $\mathcal{J}(\hat{\mathbb{Q}})/\phi(\mathcal{J}(\mathbb{Q}))$ and $\mathcal{J}(\mathbb{Q})/\hat{\phi}(\mathcal{J}(\hat{\mathbb{Q}}))$ to deduce the rank of $\mathcal{G}$. Similar to the method of complete 2 descent, one embeds both into $\mathbb{Q}^*/(\mathbb{Q}^*)^2$. Define μ^ϕ to be

$$\begin{aligned} \mu^\phi : \mathcal{J}(\mathbb{Q})/\hat{\phi}(\mathcal{J}(\hat{\mathbb{Q}})) &\longrightarrow [\mathbb{Q}^*/(\mathbb{Q}^*)^2]^{\times 2} \\ \{P, Q\} &\mapsto [G_1(x), G_2(x)] \times [G_1(u), G_2(u)] \end{aligned}$$

where P has coordinates (x, y) and Q has coordinates (u, v) .

We interpret $[G_1(x), G_2(x)]$ analogously to Lemma 1.12 when P is a K -rational Weierstrass point. If x is a root to $G_1(X)$ then $G_1(x)$ is interpreted to be $G_2(x)G_3(x)$, similarly if it is a root of $G_2(X)$ and vice versa for Q .

Analogously, define $\mu^{\hat{\phi}}$ as follows:

$$\begin{aligned} \mu^{\hat{\phi}} : \mathcal{J}(\hat{\mathbb{Q}})/\phi(\mathcal{J}(\mathbb{Q})) &\longrightarrow [\mathbb{Q}^*/(\mathbb{Q}^*)^2]^{\times 2} \\ \{P, Q\} &\mapsto [L_1(x), L_2(x)] \times [L_1(u), L_2(u)] \end{aligned}$$

where P has coordinates (x, y) and Q has coordinates (u, v) .

Lemma 3.15. 1. μ^ϕ and $\mu^{\hat{\phi}}$ are both injective group homomorphisms.

2. Let $\mathcal{S} := \{p : p \text{ prime}, p | 2\Delta b_{12}b_{13}b_{23}\hat{b}_{12}\hat{b}_{13}\hat{b}_{23}\} = \{p_1, \dots, p_m\}$ and let $\mathbb{Q}(\mathcal{S}) = \{\pm p_1^{r_1} \cdots p_m^{r_m} : r_i = 0, 1\}$. Then $\text{Im} \mu^\phi, \text{Im} \mu^{\hat{\phi}} \leq [\mathbb{Q}(\mathcal{S})]^{\times 2}$.

Proof. See [CF96, Chapter 11 and Chapter 10]. $\square$

In order to find the relevant images, a local analysis is again utilised. The relevant commutative diagrams are [Figure 3], where the maps are defined analogously to the ones in Section 1.2.2.

$$\begin{array}{ccc}
\mathcal{J}(\mathbb{Q})/\hat{\phi}(\mathcal{J}(\hat{\mathbb{Q}})) & \xrightarrow{\mu^\phi} & [\mathbb{Q}(\mathcal{S})]^{\times 2} \\
\downarrow i_p & & \downarrow j_p \\
\mathcal{J}(\mathbb{Q}_p)/\hat{\phi}(\mathcal{J}(\hat{\mathbb{Q}}_p)) & \xrightarrow{\mu_p^\phi} & [\mathbb{Q}_p^*/(\mathbb{Q}_p^*)^2]^{\times 2} \\
\mathcal{J}(\hat{\mathbb{Q}})/\phi(\mathcal{J}(\mathbb{Q})) & \xrightarrow{\mu^{\hat{\phi}}} & [\mathbb{Q}(\mathcal{S})]^{\times 2} \\
\downarrow \hat{i}_p & & \downarrow \hat{j}_p \\
\mathcal{J}(\hat{\mathbb{Q}}_p)/\phi(\mathcal{J}(\mathbb{Q}_p)) & \xrightarrow{\mu_p^{\hat{\phi}}} & [\mathbb{Q}_p^*/(\mathbb{Q}_p^*)^2]^{\times 2}
\end{array}$$

Figure 3: Richelot Isogeny - Curves of Genus 2

Now, μ_p^ϕ and $\mu_p^{\hat{\phi}}$ remain to be injective; further, $j_p^{-1}(\text{Im}\mu_p^\phi)$ and $\hat{j}_p^{-1}(\text{Im}\mu_p^{\hat{\phi}})$ provide a bound for $\text{Im}\mu^\phi$ and $\text{Im}\mu^{\hat{\phi}}$ respectively.

Since one needs to handle the cardinality of $\mathcal{J}(\mathbb{Q}_p)/\hat{\phi}(\mathcal{J}(\hat{\mathbb{Q}}_p))$ and $\mathcal{J}(\hat{\mathbb{Q}}_p)/\phi(\mathcal{J}(\mathbb{Q}_p))$, the following lemma becomes crucial during the calculations.

Lemma 3.16. Let $\mathcal{G}$ be genus 2 curve defined over $\mathbb{Q}$ and define $\mathcal{J}(\mathbb{Q}_p), \mathcal{J}(\hat{\mathbb{Q}}_p), \phi$ and $\hat{\phi}$ as above. Suppose p is a rational prime then

$$\#\mathcal{J}(\mathbb{Q}_p)/\hat{\phi}(\mathcal{J}(\hat{\mathbb{Q}}_p)) \times \#\mathcal{J}(\hat{\mathbb{Q}}_p)/\phi(\mathcal{J}(\mathbb{Q}_p)) = \begin{cases} 4 & p = \infty \\ 16 & p \neq 2, \infty \\ 64 & p = 2. \end{cases}$$

Proof. See [CF96, p. 111] for a proof. $\square$

Again, the following original example uses the machinery developed above and is inspired by Example 11.4.1 in [CF96].

Example 3.17. Let $\mathcal{G}_2 : Y^2 = G_1(X)G_2(X)G_3(X) = (X^2 + 2)(X^2 + 3)(X^2 + X)$, this has rank 1 over $\mathbb{Q}$.

Proof. Following the second example in Example 3.2, $\mathcal{J}(\mathbb{Q})[2]$ is isomorphic to $[C_2]^{\times 2}$. Let $\mathfrak{A} = \{(\sqrt{-2}, 0), (-\sqrt{-2}, 0)\}$, $\mathfrak{B} = \{(\sqrt{-3}, 0), (-\sqrt{-3}, 0)\}$ and $\mathfrak{C} = \{(0, 0), (-1, 0)\}$, then $\mathcal{J}(\mathbb{Q})[2] = \{\mathfrak{D}, \mathfrak{A}, \mathfrak{B}, \mathfrak{C}\} = \langle \mathfrak{A}, \mathfrak{B} \rangle$. Let $\mathfrak{D} = \{(0, 0), \infty^+\}$, this is an element with infinite order.

Using the theory of Richelot Isogeny, we have that $\Delta = 1$, $L_1(X) = X^2 - 6X - 3$, $L_2(X) = -X^2 + 4X + 2$, $L_3(X) = 2X$ and $b_{12} = 1$, $b_{13} = 6$, $b_{23} = 12$, $\hat{b}_{12} = 1$, $\hat{b}_{13} = -12$, $\hat{b}_{23} = 8$. Thus we have $\mathbb{Q}(\mathcal{S}) = \{\pm 1, \pm 2, \pm 3, \pm 6\}$. The isogenous curve is given by

$$\hat{\mathcal{G}} : Y^2 = (X^2 - 6X - 3)(-X^2 + 4X + 2)(2X).$$

Notice that $\mathbb{Q}(\mathcal{S}) = \langle -1, 2, 3 \rangle$ and so

$$\text{Im}\mu^\phi, \text{Im}\mu^{\hat{\phi}} \leq [\mathbb{Q}(\mathcal{S})]^{\times 2} = \langle [-1, 1], [2, 1], [3, 1], [1, -1], [1, 2], [1, 3] \rangle. \quad (3.9)$$

Magma computations on $\hat{\mathcal{G}}$ show that $\mathcal{J}(\hat{\mathbb{Q}})_{\text{tors}} = \mathcal{J}(\hat{\mathbb{Q}})[2] \cong [C_2]^{\times 2}$. Let $\hat{\mathfrak{A}} = \{(0, 0), \infty\}$, $\hat{\mathfrak{B}} = \{(3 + 2\sqrt{3}, 0), (3 - 2\sqrt{3}, 0)\}$ and $\hat{\mathfrak{C}} = \{2 + \sqrt{6}, 0\}, (2 - \sqrt{6}, 0)\}$, then $\mathcal{J}(\hat{\mathbb{Q}})_{\text{tors}} = \{\hat{\mathfrak{D}}, \hat{\mathfrak{A}}, \hat{\mathfrak{C}}, \hat{\mathfrak{B}}\} = \langle \hat{\mathfrak{A}}, \hat{\mathfrak{B}} \rangle$. Moreover, let $\hat{\mathfrak{D}} = \{(-\frac{1}{2}, \frac{1}{4}), (0, 0)\} \in \mathcal{J}(\hat{\mathbb{Q}})$, this element has infinite order.

Claim: $\text{Im}(\mu^\phi) = \langle [6, 1], [1, 3], [2, 3] \rangle$ and $\text{Im}(\mu^{\hat{\phi}}) = \langle [-3, -2], [-3, 1] \rangle$.

For simplicity, define $H = \langle [6, 1], [1, 3], [2, 3] \rangle$ and $\hat{H} = \langle [-3, -2], [-3, 1] \rangle$.

Firstly, we have the following images under μ^ϕ

$$\begin{aligned} \mathfrak{A} &\mapsto [6, 1], \\ \mathfrak{B} &\mapsto [1, 3], \\ \mathfrak{C} &\mapsto [6, 3], \\ \mathfrak{D} &\mapsto [2, 3]. \end{aligned}$$

So $H \leq \text{Im}(\mu^\phi)$.

Similarly, we have the following images under $\mu^{\hat{\phi}}$

$$\begin{aligned} \hat{\mathfrak{A}} &\mapsto [-3, -2], \\ \hat{\mathfrak{B}} &\mapsto [-3, 1], \\ \hat{\mathfrak{C}} &\mapsto [1, -2], \\ \hat{\mathfrak{D}} &\mapsto [-3, -2]. \end{aligned}$$

So $\hat{H} \leq \text{Im}(\mu^{\hat{\phi}})$.

So in order to prove the Claim, it is sufficient to show that $H = \bigcap_p j_p^{-1}(\text{Im}(\mu_p^\phi))$ and $\hat{H} = \bigcap_p j_p^{-1}(\text{Im}(\mu_p^{\hat{\phi}}))$.

For simplicity, let $\mathcal{J}_p$ and $\hat{\mathcal{J}}_p$ denote $\mathcal{J}(\mathbb{Q}_p)$ and $\mathcal{J}(\hat{\mathbb{Q}}_p)$ respectively.

Starting with $p = \infty$, by Lemma 3.16, we have that $\#\mathcal{J}_\infty/\hat{\phi}(\hat{\mathcal{J}}_\infty) \times \#\hat{\mathcal{J}}_\infty/\phi(\mathcal{J}_\infty) = 4$. Choosing a set of representatives of $\mathbb{R}^*/(\mathbb{R}^*)^2$ to be $\{\pm 1\}$, $\mathfrak{A}$, $\mathfrak{B}$ and $\mathfrak{D}$ are sent to $[1, 1]$ under μ_∞^ϕ whilst $\hat{\mathfrak{A}}$, $\hat{\mathfrak{B}}$ are sent to $[-1, -1]$, $[-1, 1]$ under $\mu_\infty^{\hat{\phi}}$ respectively.

Therefore, $\mathcal{J}_\infty/\hat{\phi}(\mathcal{J}_\infty)$ is trivial and $\hat{\mathcal{J}}_\infty/\phi(\mathcal{J}_\infty) \cong C_2 \times C_2$. In particular, $\text{Im}(\mu_\infty^\phi) = \langle [1, 1] \rangle$ whilst $\text{Im}(\mu_\infty^{\hat{\phi}}) = \langle [-1, -1], [-1, 1] \rangle$. The elements in $\mathbb{Q}(\mathcal{S})$ that reduce to the identity in $\mathbb{R}^*/(\mathbb{R}^*)^2$ is a subgroup generated by $\{2, 3\}$, this gives the following two bounds

$$\text{Im}\mu^\phi \leq j_\infty^{-1}(\text{Im}\mu_\infty^\phi) = \langle \ker j_\infty, H \rangle = \langle [2, 1], [3, 1], [1, 2], [1, 3] \rangle. \quad (3.10)$$

$$\text{Im}\mu^{\hat{\phi}} \leq \hat{j}_\infty^{-1}(\text{Im}\mu_\infty^{\hat{\phi}}) = \langle \ker \hat{j}_\infty, \hat{H} \rangle = \langle [2, 1], [3, 1], [1, 2], [1, 3], [-1, 1], [-1, -1] \rangle. \quad (3.11)$$

Notice that the second bound is a trivial one.

We repeat the process with $p = 2$ and choose a set of representatives of $\mathbb{Q}_2^*/(\mathbb{Q}_2^*)^2$ to be $\{\pm 1, \pm 2, \pm 3, \pm 6\}$. Under relevant maps we have:

$$\begin{aligned} \mathfrak{A} &\mapsto [6, 1], \\ \mathfrak{B} &\mapsto [1, 3], \\ \mathfrak{D} &\mapsto [2, 3], \\ \hat{\mathfrak{A}} &\mapsto [-3, -2], \\ \hat{\mathfrak{B}} &\mapsto [-3, 1]. \end{aligned}$$

By Lemma 3.16, there is exactly 1 generator missing for either $\mathcal{J}_2/\hat{\phi}(\hat{\mathcal{J}}_2)$ or $\hat{\mathcal{J}}_2/\phi(\mathcal{J}_2)$. After a short research, one finds $\{\{22, \gamma\}, \infty^+\} \in \mathcal{J}_2$, where $\gamma \in \mathbb{Q}_2$ is such that $\gamma^2 = 119761092$ (Such γ exists because $119761092/4 = 29940273 \equiv 1 \pmod{8}$). Let $\mathfrak{F}$ denote such point, then $\mathfrak{F} \mapsto [6, -1]$ under μ_2^ϕ .

Since $\{[6, 1], [1, 3], [2, 3], [6, -1]\}$ is a linearly independent set, $\{\mathfrak{A}, \mathfrak{B}, \mathfrak{D}, \mathfrak{F}\}$ generate $\mathcal{J}_2/\hat{\phi}(\hat{\mathcal{J}}_2)$. Similarly, $\hat{\mathfrak{A}}$ and $\hat{\mathfrak{B}}$ generate $\hat{\mathcal{J}}_2/\phi(\mathcal{J}_2)$. The elements in $\mathbb{Q}(\mathcal{S})$ that reduce to the identity in $\mathbb{Q}_2^*/(\mathbb{Q}_2^*)^2$ is the trivial subgroup and so one obtains the following bound

$$\text{Im}\mu^\phi \leq j_2^{-1}(\text{Im}\mu_2^\phi) = \langle \ker j_2, H \rangle = \langle [6, 1], [1, 3], [2, 3], [6, -1] \rangle. \quad (3.12)$$

$$\text{Im}\mu^{\hat{\phi}} \leq \hat{j}_2^{-1}(\text{Im}\mu_2^{\hat{\phi}}) = \langle \ker \hat{j}_2, \hat{H} \rangle = \langle [-3, -2], [-3, 1] \rangle. \quad (3.13)$$

(3.13) trivially shows the second part of the Claim is true.

(3.10) and (3.12) show that $H = \bigcap_p j_p^{-1}(\text{Im}(\mu_p^\phi))$. This is because (3.10) shows that the elements in $\bigcap_p j_p^{-1}(\text{Im}(\mu_p^\phi))$ have both coordinates positive while (3.12) shows that the compatible elements must be generated by $[6, 1], [1, 3]$ and $[2, 3]$. Recall that $H = \langle [6, 1], [1, 3], [2, 3] \rangle$ and $H \leq \bigcap_p j_p^{-1}(\text{Im}(\mu_p^\phi))$, one deduces $\bigcap_p j_p^{-1}(\text{Im}(\mu_p^\phi)) \leq H \leq \bigcap_p j_p^{-1}(\text{Im}(\mu_p^\phi))$ and thus the first part is also true.

Since the Claim is true, $\mathcal{J}(\mathbb{Q})/\hat{\phi}(\hat{\mathcal{J}}(\mathbb{Q})) \cong [C_2]^{\times 3}$ and $\hat{\mathcal{J}}(\mathbb{Q})/\phi(\mathcal{J}(\mathbb{Q})) \cong [C_2]^{\times 2}$, with the aid of Lemma 3.14, we obtain the following exact sequence

$$0 \longrightarrow [C_2]^{\times 2} \longrightarrow [C_2]^{\times 2} \longrightarrow \mathcal{J}(\mathbb{Q})/2\mathcal{J}(\mathbb{Q}) \cong [C_2]^{\times m} \longrightarrow [C_2]^{\times 3} \longrightarrow 0.$$

By repeatedly applying First Isomorphism Theorem if necessary, one deduces $m = 3$. Therefore the rank is 1 since $\mathcal{J}(\mathbb{Q})[2] \cong [C_2]^{\times 2}$. $\square$

4 Chabauty Techniques on Genus 2 Curves

From the last chapter, we were able to identify $\mathcal{J}(\mathbb{Q})$ for some curves of genus 2. In this chapter, we will focus on how such group could help on bounding the number of $\mathbb{Q}$ -rational points on a curve of genus 2 using Chabauty Techniques.

We will start by considering the most general case. Let $\mathcal{G}$ be a curve of genus 2 over K , assume it has the following form

$$\mathcal{G} : Y^2 = G(X) = g_6X^6 + g_5X^5 + g_4X^4 + g_3X^3 + g_2X^2 + g_1X + g_0, \quad (4.1)$$

$G(X) \in K[X]$ with $\Delta_G \neq 0$. $g_6 \neq 0$ or $g_5 \neq 0$ and $\text{char}K \neq 2, 3$ or 5 . Suppose further that $g_i \in R$, where R is a ring complete with respect to a discrete non-Archimedean valuation and has a maximal ideal $\mathcal{M}$.

4.1 Preliminary Results for a Curve of Genus 2

In this section, we will provide several elementary results regarding $\mathcal{G}$. To do so, we will closely follow Chapter 7 of [CF96].

The Jacobian of $\mathcal{G}$ is embedded into $\mathbb{P}^{15}(K)$ under the map $\mathbf{z}$ in Chapter 2 of [CF96]. Let $\mathfrak{U} \in \mathcal{J}(\mathcal{G})$ and suppose that $\mathfrak{U} \mapsto [a_0, \dots, a_{15}] \in \mathbb{P}^{15}(K)$ under $\mathbf{z}$, with $a_0 \neq 0$. Define $s_1 = a_1/a_0$ and $s_2 = a_2/a_0$ then $\mathbf{s} = (s_1, s_2)$ is a pair of local parameters for $\mathfrak{U}$, as described in Theorem 2.1 of [Fly90]. The *vector formal group*, $\mathbf{F}(\mathbf{s}, \mathbf{t}) = (F_1(\mathbf{s}, \mathbf{t}), F_2(\mathbf{s}, \mathbf{t}))$, can also be defined to serve a similar purpose to the case of elliptic curves (see Theorem 1.5 of [Fly93]). Furthermore, *Formal Logarithm*, $\mathbf{L}(\mathbf{s}) = (L_1(\mathbf{s}), L_2(\mathbf{s}))$, is defined as follows

$$\begin{aligned} L_1(s_1, s_2) &= s_1 + \frac{1}{3}(-2g_4s_1^3 + g_1s_2^3) + \text{higher order terms}, \\ L_2(s_1, s_2) &= s_2 + \frac{1}{3}(-2g_2s_2^3 + g_5s_1^3) + \text{higher order terms}, \end{aligned} \quad (4.2)$$

and that *Formal Exponential*, $\mathbf{E}(\mathbf{s}) = (E_1(\mathbf{s}), E_2(\mathbf{s}))$, is defined as follows

$$\begin{aligned} E_1(s_1, s_2) &= s_1 + \frac{1}{3}(2g_4s_1^3 - g_1s_2^3) + \text{higher order terms}, \\ E_2(s_1, s_2) &= s_2 + \frac{1}{3}(2g_2s_2^3 - g_5s_1^3) + \text{higher order terms}, \end{aligned} \quad (4.3)$$

satisfying $\mathbf{L}(\mathbf{F}(\mathbf{s}, \mathbf{t})) = \mathbf{L}(\mathbf{s}) + \mathbf{L}(\mathbf{t})$ and $\mathbf{E}(\mathbf{s} + \mathbf{t}) = \mathbf{F}(\mathbf{E}(\mathbf{s}), \mathbf{E}(\mathbf{t}))$, where $\mathbf{s}, \mathbf{t}$ are local parameters for two points on $\mathcal{G}$. (See Section 7.2 of [CF96] for a detailed derivation.)

4.2 Classical Chabauty Techniques

Suppose now $\mathcal{G}$ is defined over $\mathbb{Q}$ and possesses the form in (4.1); in other words, $g_1, \dots, g_6 \in \mathbb{Z}$. Suppose further that the Jacobian has rank 1, since the case of rank 0 means there are only finitely many points one has to check, by Theorem 3.3. This section will be dedicated to bounding the number of $\mathbb{Q}$ -rational points on $\mathcal{G}$ using classical Chabauty, where the ideas will be very similar to the ones in Section 2.2. To do so, we will follow Chapter 13 of [CF96] and Chapter 3 of [Fly00].

Firstly, a finite check on the roots of $G(X)$ gives a set of $\mathbb{Q}$ -rational Weierstrass points on $\mathcal{G}$. So suppose $\mathfrak{X} \in \mathcal{G}(\mathbb{Q})$ is not a Weierstrass point, one could map this point to $\{\mathfrak{X}, \mathfrak{X}\} \in \mathcal{J}(\mathbb{Q})$. Such map is injective since $\mathfrak{X}$ is assumed to be a non-Weierstrass point. Moreover, any $\{\mathfrak{Y}, \mathfrak{Y}\} \in \mathcal{J}(\mathbb{Q})$ corresponds to $\mathfrak{Y} \in \mathcal{G}(\mathbb{Q})$. Thus it is then sufficient to find all $\{\mathfrak{X}, \mathfrak{X}\} \in \mathcal{J}(\mathbb{Q})$.

Since $\mathcal{G}$ has rank 1, suppose $\mathcal{J}(\mathbb{Q}) = \langle \mathcal{J}(\mathbb{Q})_{\text{tors}}, P \rangle$ where P is the generator of the torsion free part. Let p be a rational prime and a good reduction on $\mathcal{G}$ and let $\tilde{\mathcal{J}}(\mathbb{F}_p)$ and $\tilde{P}$ denote the modulo p reduction of $\mathcal{J}(\mathbb{Q})$ and P respectively. Define m to be the order of $\tilde{P}$ in $\tilde{\mathcal{J}}(\mathbb{F}_p)$ and so $Q = mP$ lies in kernel of the reduction map $\mathcal{J}(\mathbb{Q}) \rightarrow \tilde{\mathcal{J}}(\mathbb{F}_p)$. Let $\mathcal{T}$ be the following finite set:

$$\{S + lP : S \in \mathcal{J}(\mathbb{Q})_{\text{tors}}, \lfloor -m/2 \rfloor + 1 \leq l \leq \lfloor m/2 \rfloor\}, \quad (4.4)$$

so that every $L \in \mathcal{J}(\mathbb{Q})$ could be uniquely written as $L = T + nQ$, $T \in \mathcal{T}$, $n \in \mathbb{Z}$. One may compute the local parameter $\mathbf{s}$ of Q using Equation (26) in [Fly00] and thus the local parameter $\mathbf{t} = \mathbf{t}(\mathbf{n})$ of nQ by $\mathbf{E}(n\mathbf{L}(\mathbf{s}))$.

Suppose $T + nQ = \{(X, Y), (U, V)\}$ and let $\mathbf{t}$ be as above. The theory of Kummer Surface⁸ gives the following equation as projective coordinates

$$[1 : X + U, XU] = [\phi_1(\mathbf{t}, T), \phi_2(\mathbf{t}, T), \phi_3(\mathbf{t}, T)], \quad (4.5)$$

where $\phi_i(\mathbf{t}, T)$ is merely a power series in n , depending on $\mathbf{z}(T)$, with coefficients in $\mathbb{Z}_p$. Suppose n is such that $(X, Y) = (U, V)$, one therefore requires $(X+U)^2 - 4XU = 0$; this means that n is a solution to the power series $\phi_2(\mathbf{t}, T)^2 - 4\phi_1(\mathbf{t}, T)\phi_3(\mathbf{t}, T)$.

With Q chosen as above, define $\sigma: \mathcal{T} \rightarrow \mathbb{Z}_p[[n]]$:

$$\sigma(T) = \phi_2(\mathbf{t}, T)^2 - 4\phi_1(\mathbf{t}, T)\phi_3(\mathbf{t}, T). \quad (4.6)$$

Hence, it is now sufficient to bound the solutions of $\sigma(T)$ for each $T \in \mathcal{T}$, using Strassman's Theorem again. Summing over all possible $T \in \mathcal{T}$, this gives an overall bound of the number of $\mathbb{Q}$ -rational points on $\mathcal{G}$.

4.3 Worked Example

We are now in the position to apply the above tools, the following original example is inspired by Example 13.2.1 of [CF96].

⁸See Chapter 3 and 13 of [CF96].

Example 4.1. An upper bound of $\mathbb{Q}$ -rational points on $\hat{\mathcal{G}} : Y^2 = (X^2 - 6X - 3)(-X^2 + 4X + 2)(2X)$ is 10. In particular, there are 4 known $\mathbb{Q}$ -rational points, namely $\infty, (0, 0), (-\frac{1}{2}, -\frac{1}{4})$ and $(-\frac{1}{2}, \frac{1}{4})$.

Proof. Firstly, a search of the Weierstrass points on $\hat{\mathcal{G}}$ reveals two $\mathbb{Q}$ -rational points, ∞ and $(0, 0)$.

Let $\mathcal{J}(\mathbb{Q})$ denote the Mordell-Weil group of $\hat{\mathcal{G}}$. Recall in Example 3.17, $\hat{\mathcal{G}}(\mathbb{Q})_{\text{tors}} = \{\hat{\mathfrak{D}}, \hat{\mathfrak{A}}, \hat{\mathfrak{B}}, \hat{\mathfrak{C}}\}$. Moreover, let $\hat{\mathfrak{C}} = \{(-\frac{1}{2}, \frac{1}{4}), \infty\}$, this again has infinite order. The bad primes in this case are, unsurprisingly, 2 and 3 and so we start applying Chabauty techniques with $p = 5$.

Let $\tilde{\mathcal{G}}, \tilde{J}(\mathbb{F}_5)$ and $\tilde{E}$ be the reduction of $\hat{\mathcal{G}}, \mathcal{J}(\mathbb{Q})$ and $\hat{\mathfrak{C}}$ modulo 5 respectively. Magma shows that $\tilde{J}(\mathbb{F}_5) \cong \mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/10$ and that $\tilde{E} \in \tilde{J}(\mathbb{F}_5)$ has order 10. Adapting the idea in Comment 2.4, it is sufficient to consider $\hat{\mathfrak{C}}$ as a generator of the torsion free part for this calculation. This is because suppose $\hat{\mathfrak{C}}_0$ is a generator of the torsion free part then $\exists W \in \mathbb{Z}$ such that $\hat{\mathfrak{C}} = W\hat{\mathfrak{C}}_0$. We proceed by showing that $5 \nmid W$. If this is not true then $10 \cdot (\frac{W}{5})\hat{\mathfrak{C}}_0 = 2W\hat{\mathfrak{C}}_0 = 2\hat{\mathfrak{C}}$ is in the kernel of reduction, a contradiction. Alternatively, a finite field argument using mod 13 achieves the same. Similarly one could show $2 \nmid W$. Thus any element $L \in \mathcal{J}(\mathbb{Q})$ can be written as $L = L' + n\hat{\mathfrak{C}}, L' \in \mathcal{J}(\mathbb{Q})_{\text{tors}}, n \in \mathbb{Z}_5$.

Using the previous notations, let $Q = 10\hat{\mathfrak{C}}$, this does not possess the required form. Let $\mathcal{T}$ be the following finite set

$$\mathcal{T} = \{S + n\hat{\mathfrak{C}} : S \in \mathcal{J}(\mathbb{Q})_{\text{tors}}, -4 \leq n \leq 5\}, \quad (4.7)$$

so that any $L \in \mathcal{J}(\mathbb{Q})$ can be written as $L = T + nQ, T \in \mathcal{T}, n \in \mathbb{Z}_5$. Since we require L to be in the form of $\{P, P\}$, it is sufficient to only consider the elements in $\mathcal{T}$ such that it reduces to such form under the reduction map $\mathcal{J}(\mathbb{Q}) \rightarrow \tilde{J}(\mathbb{F}_5)$. A finite check with Magma shows that the only compatible elements are $\hat{\mathfrak{D}}, \pm 2\hat{\mathfrak{C}}, \pm 4\hat{\mathfrak{C}}$ and indeed $\mathcal{T}$ does not contain any element in the kernel of reduction. Using Equation (26) in [Fly00], we can calculate that the local parameters of Q . With the aid of SageMath, this gives $\mathbf{s} = (s_1, s_2) = (580, 440) \pmod{5^4}$. By (4.2), $\mathbf{L}(\mathbf{s}) = (205, 440) \pmod{5^4}$. Then using (4.3), the local parameter of nQ is $\mathbf{t} = \mathbf{t}(\mathbf{n}) = (t(n)_1, t(n)_2) = (205n + 375n^3, 440n) \pmod{5^4}$.

We first consider $\hat{\mathfrak{D}}$ to obtain an upper bound on the possible n such that nQ has the form $\{\mathfrak{X}, \mathfrak{X}\}$. We proceed by using Appendix B of [Fly90] to calculate $\mathbf{z}(nQ)$ and using **jacobian.variety** directory⁹ to calculate $\phi_i(\mathbf{t}, T)$. This is now sufficient to calculate $\sigma(\hat{\mathfrak{D}})$, following the procedure in Chapter 13 of [CF96]. Notice that $\mathbf{z}(\hat{\mathfrak{D}}) = [1, 0, \dots, 0] \in \mathbb{P}^{15}$ and so a short calculation reveals that $\sigma(\hat{\mathfrak{D}})$ is a power series with only even degrees and

$$\sigma(\hat{\mathfrak{D}}) = O(n^6) = 15625n^6 \pmod{5^7}. \quad (4.8)$$

⁹See <https://people.maths.ox.ac.uk/flynn/genus2/jacobian.variety/bilinear.forms>.

Notice that $|15625|_5 = 5^{-6}$ and by Strassman's Theorem to $\sigma(\hat{\mathfrak{D}})/n^6$, this gives $n = 0$ as the only root, which just returns $\hat{\mathfrak{D}}$.

Next, we consider $\pm 2\hat{\mathfrak{E}}$. Notice that it is sufficient to only consider $2\hat{\mathfrak{E}}$ because if there exists n such that $2\hat{\mathfrak{E}} + nQ$ has the required form then $-2\hat{\mathfrak{E}} - nQ$ has the same form as well, and vice versa. Using Magma, we first recognise that $2\hat{\mathfrak{E}} = \{(-\frac{1}{2}, \frac{1}{4}), (-\frac{1}{2}, -\frac{1}{4})\}$ and so this indicates that $n = 0$ will be a solution. Moreover, a short calculation shows that

$$\begin{aligned}\phi_1(\mathbf{t}, 2\hat{\mathfrak{E}}) &= 362 + 50n + 225n^2 + 125n^3 \pmod{5^4}, \\ \phi_2(\mathbf{t}, 2\hat{\mathfrak{E}}) &= 263 + 615n + 275n^2 + 500n^3 \pmod{5^4}, \\ \phi_3(\mathbf{t}, 2\hat{\mathfrak{E}}) &= 403 + 130n + 50n^2 + 500n^3 \pmod{5^4}.\end{aligned}\tag{4.9}$$

This means that

$$\begin{aligned}\sigma(2\hat{\mathfrak{E}}) &= \phi_2(\mathbf{t}, 2\hat{\mathfrak{E}})^2 - 4\phi_1(\mathbf{t}, 2\hat{\mathfrak{E}})\phi_3(\mathbf{t}, 2\hat{\mathfrak{E}}) = O(n) \\ &= 275n + 525n^2 \pmod{5^4}.\end{aligned}\tag{4.10}$$

Notice that $|275|_5 = |525|_5 = \frac{1}{25}$ and so we could apply Strassman's Theorem to deduce that there are at most 2 roots, one of which is $n = 0$.

Next, we consider $-2\hat{\mathfrak{E}}$, notice that $-2\hat{\mathfrak{E}} = \{(-\frac{1}{2}, -\frac{1}{4}), (-\frac{1}{2}, \frac{1}{4})\}$. As stated above, there is no need to carry out calculations for $-2\hat{\mathfrak{E}}$ because there are at most 2 roots to $\sigma(-2\hat{\mathfrak{E}})$ with one being $n = 0$.

It now remains to carry out the same calculation with $\pm 4\hat{\mathfrak{E}}$. Again, it is sufficient to consider $4\hat{\mathfrak{E}}$ only. Firstly, Magma shows that $4\hat{\mathfrak{E}} = \{P, \hat{P}\}$, $P = (\frac{1}{249218}(12\beta - 97203), -\frac{1}{21924692867972}(5447142915\beta + 47851326827886))$, where $\beta = \sqrt{-116696705}$ and $\hat{P}$ is the conjugate of P . Notice that this element does not possess the required form. As usual we proceed by computing $\sigma(4\hat{\mathfrak{E}})$. A short calculation reveals

$$\begin{aligned}\phi_1(\mathbf{t}, 4\hat{\mathfrak{E}}) &= 135 + 110n + 250n^2 + 375n^3 \pmod{5^4}, \\ \phi_2(\mathbf{t}, 4\hat{\mathfrak{E}}) &= 580 + 570n + 600n^2 + 500n^3 \pmod{5^4}, \\ \phi_3(\mathbf{t}, 4\hat{\mathfrak{E}}) &= 585 + 290n + 100n^2 \pmod{5^4}.\end{aligned}\tag{4.11}$$

So we have

$$\begin{aligned}\sigma(4\hat{\mathfrak{E}}) &= \phi_2(\mathbf{t}, 4\hat{\mathfrak{E}})^2 - 4\phi_1(\mathbf{t}, 4\hat{\mathfrak{E}})\phi_3(\mathbf{t}, 4\hat{\mathfrak{E}}) = O(1) \\ &= 125 + 325n + 550n^2 \pmod{5^4}.\end{aligned}\tag{4.12}$$

For this case, $|125|_5 = \frac{1}{125}$, $|325|_5 = \frac{1}{25}$ and $|550|_5 = \frac{1}{25}$; by Strassman's Theorem there are at most 2 solutions to $\sigma(4\hat{\mathfrak{E}})$. Therefore, using the observation above, there are at most 2 solutions to $\sigma(-4\hat{\mathfrak{E}})$, too.

Thus, after summing over all $T \in \mathcal{T}$, there are at most 10 $\mathbb{Q}$ -rational points on $\hat{\mathcal{G}}$. There are 4 known $\mathbb{Q}$ -rational points; two from the analysis on Weierstrass points, namely ∞ and $(0, 0)$, and two from the analysis on $2\hat{\mathfrak{E}}$, namely $(-\frac{1}{2}, -\frac{1}{4})$ and $(-\frac{1}{2}, \frac{1}{4})$. $\square$

The same bound could be produced using Coleman's Theorem.

Theorem 4.2 (Coleman's Theorem). [CF96, p. 152] Let $\mathcal{G}$ be a curve of genus 2 over $\mathbb{Q}$ with rank at most 1 and suppose $p \geq 5$ is a prime of good reduction then $\#\mathcal{G}(\mathbb{Q}) \leq \#\hat{\mathcal{G}}(\mathbb{F}_p) + 2$.

Comment 4.3. From the above, we have seen that $p = 5$ is a prime of good reduction. Let $\hat{\mathcal{G}}'$ be the reduction modulo 5. Magma shows that $\#\hat{\mathcal{G}}' = 8$. Thus Coleman's Theorem gives an upper bound of 10, again.

5 Covering of Bielliptic Curves

Suppose $\mathcal{G}$ is a curve of genus 2 with rank greater than 1 then the method of Chabauty described in Chapter 4 cannot be applied directly. In this Chapter, we will make an attempt to solve some of these problems. In particular, we will focus on *bielliptic curves* over K .

Definition 5.1. [Wet97, p. 23] A curve over K is said to be a **bielliptic curve** if it has a degree¹⁰ 2 map to an elliptic curve.

While this definition only provides an abstract concept, the following theorem from Section 4.1 of [Wet97] gives a concrete description of such curves:

Theorem 5.2. Let $\mathcal{G}$ be a bielliptic curve over K , with a K -rational point, then

1. $\exists b(X) = b_0 + b_1X + b_2X^2 + b_3X^3 \in K[X]$ such that

$$\mathcal{G} : Y^2 = b(X^2) = b_0 + b_1X^2 + b_2X^4 + b_3X^6.$$

2. Suppose $\Delta_b \neq 0$. Define

$$\mathcal{C}_1 : Y^2 = b(X) = b_0 + b_1X + b_2X^2 + b_3X^3$$

and

$$\mathcal{C}_2 : Y^2 = X^3b(1/X) = b_0X^3 + b_1X^2 + b_2X + b_3,$$

then we have maps $\phi_1 : \mathcal{G} \rightarrow \mathcal{C}_1$ via $\phi_1(x, y) = (x^2, y)$ and $\phi_2 : \mathcal{G} \rightarrow \mathcal{C}_2$ via $\phi_2(x, y) = (1/x^2, y/x^3)$.

3. ϕ_1 and ϕ_2 induce isogenies between $\mathcal{C}_1(K) \times \mathcal{C}_2(K)$ and $\mathcal{J}(K)$. In particular, $\exists \rho_1 : \mathcal{C}_1(K) \times \mathcal{C}_2(K) \rightarrow \mathcal{J}(K)$ and $\exists \rho_2 : \mathcal{J}(K) \rightarrow \mathcal{C}_1(K) \times \mathcal{C}_2(K)$ ¹¹ with the following diagram being commutative:

¹⁰The notion of *degree* is defined in C3.3 Differentiable Manifold Lecture Notes, Section 7.2. Available at https://courses.maths.ox.ac.uk/pluginfile.php/12530/mod_resource/content/2/dm2122.pdf.

¹¹ ρ_i are defined explicitly in [Fly00], Equation 37.

$$\begin{array}{ccc} \mathcal{C}_1(K) \times \mathcal{C}_2(K) & \xrightarrow{\rho_1} \mathcal{J}(K) & \xrightarrow{\rho_2} \mathcal{C}_1(K) \times \mathcal{C}_2(K). \\ & \searrow & \nearrow \\ & [2]_{\mathcal{C}_1(K) \times \mathcal{C}_2(K)} & \end{array}$$

4. $\ker \rho_1 \cong [C_2]^{\times 2}$ and $\ker \rho_2 \cong [C_2]^{\times 2}$.

Comment 5.3. Since the kernels are finite, the sum of the ranks of $\mathcal{C}_1$ and $\mathcal{C}_2$ is the same as the rank of $\mathcal{G}$.

Suppose $K = \mathbb{Q}$ and that $b(X) = (X - \alpha)(X - \beta)(X - \gamma)$ over some number field L . We will follow [Fly00, Chapter 4] in the pursuit of bounding the number of $\mathbb{Q}$ -rational points on $\mathcal{G}$. Let $L_1 = \mathbb{Q}(\alpha)$, $L_2 = \mathbb{Q}(\beta)$ and $L_3 = \mathbb{Q}(\gamma)$; first consider the following injective group homomorphism

$$\begin{aligned} \xi: \mathcal{J}(\mathbb{Q})/\rho_1(\mathcal{C}_1(\mathbb{Q}) \times \mathcal{C}_2(\mathbb{Q})) &\longrightarrow [L_1^*/(L_1^*)^2] \times [L_2^*/(L_2^*)^2] \times [L_3^*/(L_3^*)^2] \\ \{(x, y), (u, v)\} &\mapsto [x^2 - \alpha, x^2 - \beta, x^2 - \gamma] \times [u^2 - \alpha, u^2 - \beta, u^2 - \gamma]. \end{aligned}$$

Suppose $\mathcal{J}(\mathbb{Q})/\rho_1(\mathcal{C}_1(\mathbb{Q}) \times \mathcal{C}_2(\mathbb{Q})) = \{R_1, \dots, R_n\}$, for some $R_i \in \mathcal{J}(\mathbb{Q})$; such set can be found using descent. Let $(x, y) \in \mathbb{Q}(\mathcal{G})$ and consider $\{(x, y), \infty^+\} \in \mathcal{J}(\mathbb{Q})$, then $\{(x, y), \infty^+\} = R_j$ in $\mathcal{J}(\mathbb{Q})/\rho_1(\mathcal{C}_1(\mathbb{Q}) \times \mathcal{C}_2(\mathbb{Q}))$ for some $j \in \{1, \dots, n\}$.

Let $\xi(R_j) = [R_j^{(1)}, R_j^{(2)}, R_j^{(3)}]$ then $R_j^{(1)} = x^2 - \alpha$ in $L_1^*/(L_1^*)^2$, $R_j^{(2)} = x^2 - \beta$ in $L_2^*/(L_2^*)^2$ and $R_j^{(3)} = x^2 - \gamma$ in $L_3^*/(L_3^*)^2$. We first focus on $R_j^{(1)} = x^2 - \alpha$ in $L_1^*/(L_1^*)^2$; this means that $\exists \tilde{y} \in L_1^*$ such that

$$\tilde{y}^2(x^2 - \alpha) = R_j^{(1)}.$$

Recall that $b(x^2)$ is also square and so $\exists \hat{y} \in L_1^*$ such that

$$\hat{y}^2 = R_j^{(1)}b(x^2)/(x^2 - \alpha).$$

This is now a curve with genus 1 with the right hand side having degree 4, a further transformation $X = x^2$ and $Y = x\hat{y}$ gives the following elliptic curve over L_1

$$\mathcal{F}_{1,j} : Y^2 = R_j^{(1)}b(x)x/(x - \alpha),$$

this now has the standard form. One can analogously produce $\mathcal{F}_{2,j}$ and $\mathcal{F}_{3,j}$ by considering $R_j^{(2)}$ and $R_j^{(3)}$. Thus, $(x, y) \in \mathbb{Q}(\mathcal{G})$ has the property that $X = x^2$ is the x -coordinate of a point on $\mathcal{F}_{i,j}$ for each i and so we may be able to apply Elliptic Curve Chabauty described in Chapter 2. Repeating this process with other R_l gives an overall bound. If the rank of $\mathcal{G}$ is greater than 1, this method potentially works even though classical Chabauty from Chapter 4 is no longer applicable.

VI.17 of the *Arithmetica* concerns about positive rational solutions to the equation $y^2 = x^8 + x^4 + x^2$. Trivially, $(0, 0)$ is a solution and if $x \neq 0$ let $X = x$ and $Y = y/x$ to obtain the bielliptic curve

$$\mathcal{G}_3 : Y = X^6 + X^2 + 1.$$

Thus it is sufficient just to determine the rational solutions on $\mathcal{G}_3$. Using Theorem 5.2 and Magma, one deduces that $\mathcal{G}_3$ has rank 2 since both $\mathcal{C}_{3,1} : Y = X^3 + X + 1$ and $\mathcal{C}_{3,2} : Y = X^3 + X^2 + 1$ have rank 1 over $\mathbb{Q}$, thus classical Chabauty cannot be applied. Indeed, Theorem 7 of [Fly00] resolves this issue using the above method.

This motivates the following original example, although there is a quicker approach via resultants in [Fly00, Chapter 6].

Example 5.4. Let $\mathcal{G}_4 : Y^2 = (X^4 + 1)(X^2 + 1)$, $\mathcal{G}_4(\mathbb{Q}) = \{\infty^\pm, (0, \pm 1), (\pm 1, \pm 2)\}$.

Proof. Using the same notation, $b(X) = (X + i)(X - i)(X + 1)$ over $\mathbb{Q}(i)$ and $\mathcal{C}_{4,1} = \mathcal{C}_{4,2} : Y^2 = X^3 + X^2 + X + 1$, this elliptic curve has rank 1 over $\mathbb{Q}$. Moreover, Magma shows that $\mathcal{J}(\mathbb{Q})_{\text{tors}} = \{\mathcal{O}, \{(i, 0), (-i, 0)\}\}$ and so $\mathcal{J}(\mathbb{Q}) \cong \mathbb{Z}/2 \times \mathbb{Z}^2$. Thus, under $\xi : \mathcal{J}(\mathbb{Q})/\rho_1(\mathcal{C}_{4,1}(\mathbb{Q}) \times \mathcal{C}_{4,2}(\mathbb{Q})) \longrightarrow [\mathbb{Q}(i)^*/(\mathbb{Q}(i)^*)^2] \times [\mathbb{Q}(i)^*/(\mathbb{Q}(i)^*)^2] \times [\mathbb{Q}^*/(\mathbb{Q}^*)^2]$ we have

$$\begin{aligned} \{(i, 0), (-i, 0)\} &\mapsto [1, 1, 1], \\ \{(-1, 2), \infty^+\} &\mapsto [1 + i, 1 - i, 2], \\ \{(0, 1), \infty^+\} &\mapsto [i, i, 1], \\ \{(-1, 2), (0, 1)\} &\mapsto [i(1 + i), i(1 - i), 2]. \end{aligned}$$

Since the rank is 2 and $\mathcal{J}(\mathbb{Q})_{\text{tors}}$ is mapped to identity under ξ , we deduce that

$$\mathcal{J}(\mathbb{Q})/\rho_1(\mathcal{C}_{4,1}(\mathbb{Q}) \times \mathcal{C}_{4,2}(\mathbb{Q})) = \{\mathcal{O}, \{(-1, 2), \infty^+\}, \{(0, 1), \infty^+\}, \{(-1, 2), (0, 1)\}\}.$$

We could in fact avoid the use of Elliptic Curve Chabauty by considering the last coordinate of $\text{Im}(\xi)$; one produces the following elliptic curves over $\mathbb{Q}$

$$\begin{aligned} \mathcal{F}_{3,1} : Y^2 &= X(X^2 + 1), \\ \mathcal{F}_{3,2} : Y^2 &= 2X(X^2 + 1), \\ \mathcal{F}_{3,3} : Y^2 &= X(X^2 + 1), \\ \mathcal{F}_{3,4} : Y^2 &= 2X(X^2 + 1). \end{aligned}$$

Thus we consider $\mathcal{F}_{3,1} : Y^2 = X(X^2 + 1)$ and $\mathcal{F}_{3,2} : Y^2 = 2X(X^2 + 1)$. With a bit of luck, Magma shows that both of these curves have rank 0; $\mathcal{F}_{3,1}(\mathbb{Q}) \cong \mathbb{Z}/2$ and $\mathcal{F}_{3,2}(\mathbb{Q}) \cong \mathbb{Z}/4$. Thus it is sufficient to consider $(x_1, y_1) \in \mathcal{F}_{3,1}(\mathbb{Q})$ and $(x_2, y_2) \in \mathcal{F}_{3,2}(\mathbb{Q})$ such that $x_i \in (\mathbb{Q})^2$. A quick search shows $\mathcal{F}_{3,1}(\mathbb{Q}) = \{\infty, (0, 0)\}$ and $\mathcal{F}_{3,2}(\mathbb{Q}) = \{\infty, (1, 2), (0, 0), (1, -2)\}$. Thus if $(x, y) \in \mathcal{G}_4(\mathbb{Q})$ then $x^2 = 0$ or 1. Another quick calculation shows indeed $x = 0, x = 1, x = -1$ has corresponding y points. Thus, we conclude $\mathcal{G}_4(\mathbb{Q}) = \{\infty^\pm, (0, \pm 1), (\pm 1, \pm 2)\}$. $\square$

Comment 5.5. Alternatively, one could also consider the following elliptic curves

over $\mathbb{Q}(i)$ by considering the first coordinate of $\text{Im}(\xi)$

$$\begin{aligned}\mathcal{F}_{1,1} : Y^2 &= X(X-i)(X+1), \\ \mathcal{F}_{1,2} : Y^2 &= (1+i)X(X-i)(X+1), \\ \mathcal{F}_{1,3} : Y^2 &= iX(X-i)(X+1), \\ \mathcal{F}_{1,4} : Y^2 &= i(1+i)X(X-i)(X+1).\end{aligned}$$

Magma shows that $\mathcal{F}_{1,1}(\mathbb{Q}(i)) \cong \mathbb{Z}/2 \times \mathbb{Z}/2$, $\mathcal{F}_{1,2}(\mathbb{Q}(i)) \cong \mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}$, $\mathcal{F}_{1,3}(\mathbb{Q}(i)) \cong \mathbb{Z}/2 \times \mathbb{Z}/2$ and $\mathcal{F}_{1,4}(\mathbb{Q}(i)) \cong \mathbb{Z}/2 \times \mathbb{Z}/2$. Thus it is sufficient to only carry out Elliptic Curve Chabauty on $\mathcal{F}_{1,2} : Y^2 = (1+i)X(X-i)(X+1)$ to produce a bound since other curves have rank 0 over $\mathbb{Q}(i)$.

6 Appendix

6.1 Completion of a Number Field at $\mathfrak{p}$

Let K be a number field and $\mathcal{O}$ be the ring of integers of K . Suppose $\mathfrak{p}$ is a prime ideal of $\mathcal{O}$. $\mathcal{O}/\mathfrak{p}$ is a finite field and so define $\text{char}(\mathfrak{p})$ to be the characteristic of $\mathcal{O}/\mathfrak{p}$.

Suppose we have a normalised discrete valuation $\nu_{\mathfrak{p}} : K^* \rightarrow \mathbb{Z}$ (in other words, a surjective valuation) associated to $\mathfrak{p}$, we can then define the following metric $|\cdot|_{\mathfrak{p}}$ on K :

$$|x|_{\mathfrak{p}} = \begin{cases} N(\mathfrak{p})^{-\nu_{\mathfrak{p}}(x)} & x \in K^* \\ 0 & x = 0, \end{cases}$$

where we recall $N(\mathfrak{p})$ is the norm of the ideal $\mathfrak{p}$.

Then $K_{\mathfrak{p}}$ is the metric completion of K with respect to $|x|_{\mathfrak{p}}$. We can further define $\mathcal{O}_{\mathfrak{p}} = \{x \in K_{\mathfrak{p}} : \nu_{\mathfrak{p}}(x) \geq 0\}$ and $\mathfrak{p}\mathcal{O}_{\mathfrak{p}} = \{x \in K_{\mathfrak{p}} : \nu_{\mathfrak{p}}(x) \geq 1\}$, the latter is a maximal ideal of $\mathcal{O}_{\mathfrak{p}}$. ([Bru99, Chapter 2])

Comment 6.1. This generalises the case $K = \mathbb{Q}$; let $\mathfrak{p}$ be any rational prime and define

$$\begin{aligned}\tilde{\nu}_{\mathfrak{p}} : \mathbb{Z} \setminus \{0\} &\longrightarrow \mathbb{Z}^{\geq 0} \\ N &\mapsto \text{the highest power of } \mathfrak{p} \text{ that divides } N.\end{aligned}$$

Then $\tilde{\nu}_{\mathfrak{p}}$ extends its domain to $\mathbb{Q}^*$ as follows:

$$\begin{aligned}\nu_{\mathfrak{p}} : \mathbb{Q}^* &\longrightarrow \mathbb{Z} \\ \frac{a}{b} &\mapsto \tilde{\nu}_{\mathfrak{p}}(a) - \tilde{\nu}_{\mathfrak{p}}(b).\end{aligned}$$

One may see that this recovers the usual $\mathfrak{p}$ -adic valuation.

6.2 Non-singular model of curves of Genus 2 in $\mathbb{P}^4$

Recall that the form of a hyperelliptic curve over K was taken to be

$$\mathcal{G} : Y^2 = g_6 X^6 + g_5 X^5 + g_4 X^4 + g_3 X^3 + g_2 X^2 + g_1 X + g_0.$$

However, the points at infinity in the projective closure of this curve are singular. To 'de-singularise' this model, one can produce the following in $\mathbb{P}^4(K)$ ([CF96, Chapter 1]):

$$\begin{aligned} Y^2 &= g_6 X_3^2 + g_5 X_2 X_3 + g_4 X_2^2 + g_3 X_1 X_2 + g_2 X_1^2 + g_1 X_0 X_1 + g_0 X_0^2, \\ X_0 X_2 - X_1^2 &= 0, \\ X_0 X_3 - X_1 X_2 &= 0, \\ X_1 X_3 - X_2^2 &= 0, \end{aligned}$$

where X_i is interpreted as X^i . Thus the points on this model are in the form of $[X_0, X_1, X_2, X_3, Y]$ and an affine point (x, y) is identified as $[1, x, x^2, x^3, y]$.

Take $[x_0, x_1, x_2, x_3, y]$ to be a point of the above model. If $x_0 = 0$, the above relations suggest that $x_1 = x_2 = 0$ and so the equation reduces to $Y^2 = g_6 X_3^2$. If $x_3 = 0$ then so would be y and thus $x_3 \neq 0$; normalise so that $x_3 = 1$ and so $y = \pm\sqrt{g_6}$. We thus produced two points at infinity if $g_6 \neq 0$:

$$\begin{aligned} \infty^+ &= [0, 0, 0, 1, \sqrt{g_6}], \\ \infty^- &= [0, 0, 0, 1, -\sqrt{g_6}]. \end{aligned}$$

If $g_6 = 0$ then there is only one point at infinity, namely

$$\infty = [0, 0, 0, 1, 0].$$

([Lan, Chapter 5])

With such explanation, it is obvious that when $g_6 = 0$, the point at infinity is seen as K -rational and is a Weierstrass point; when $g_6 \neq 0$ then the points at infinity are seen to be K -rational if and only if $g_6 \in (K^*)^2$, otherwise the points at infinity are not K -rational and perceived to be conjugates of each other. However, they are not Weierstrass points.

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